

145. Consider the equation $x^3 - ax^2 + bx - c = 0$, where a, b, c are rational number, $a \neq 1$. It is given that x_1, x_2 and $x_1 x_2$ are the real roots of the equation. Then $x_1 x_2 \left(\frac{a+1}{b+c} \right) =$
- (a) 1 (b) 2 (c) 3 (d) 4
146. The exhaustive set of values of a for which inequation $(a-1)x^2 - (a+1)x + a - 1 \geq 0$ is true $\forall x \geq 2$.
- (a) $(-\infty, 1)$ (b) $\left[\frac{7}{3}, \infty \right)$ (c) $\left[\frac{3}{7}, \infty \right)$ (d) None of these
147. The number of real solutions of the equation $x^2 - 3|x| + 2 = 0$
- (a) 2 (b) 4 (c) 1 (d) 3
148. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has
- (a) infinite number of real roots (b) no real root
(c) exactly one real root (d) exactly four real roots
149. If α, β are the roots of the quadratic equation $x^2 - 2(1 - \sin 2\theta)x - 2\cos^2 2\theta = 0$, ($\theta \in R$) then the minimum value of $(\alpha^2 + \beta^2)$ is equal to :
- (a) -4 (b) 8 (c) 0 (d) 2
150. If the equation $|\sin x|^2 + |\sin x| + b = 0$ has two distinct roots in $[0, \pi]$; then the number of integers in the range of b is equals to :
- (a) 0 (b) 1 (c) 2 (d) 3
151. If $a \neq 0$ and the equation $ax^2 + bx + c = 0$ has two roots α and β such that $\alpha < -3$ and $\beta > 2$. Which of the following is always true ?
- (a) $a(a + |b| + c) > 0$ (b) $a(a + |b| + c) < 0$
(c) $9a - 3b + c > 0$ (d) $(9a - 3b + c)(4a + 2b + c) < 0$
152. If α, β are the roots of the quadratic equation $x^2 + px + q = 0$ and γ, δ are the roots of $x^2 + px - r = 0$ then $(\alpha - \gamma)(\alpha - \delta)$ is equal to :
- (a) $q + r$ (b) $q - r$ (c) $-(q + r)$ (d) $-(p + q + r)$
153. Complete set of solution of $\log_{1/3}(2^{x+2} - 4^x) \geq -2$ is :
- (a) $(-\infty, 2)$ (b) $(-\infty, 2 + \sqrt{13})$ (c) $(2, \infty)$ (d) None of these

Answers

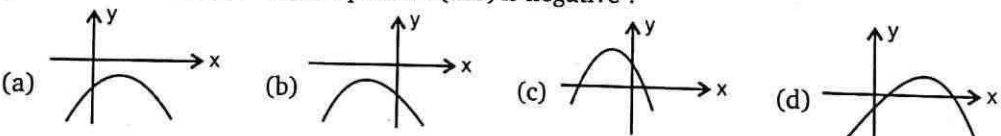
1. (d)	2. (c)	3. (a)	4. (b)	5. (a)	6. (a)	7. (b)	8. (c)	9. (a)	10. (d)
11. (c)	12. (c)	13. (a)	14. (a)	15. (d)	16. (d)	17. (b)	18. (b)	19. (c)	20. (a)
21. (b)	22. (d)	23. (c)	24. (a)	25. (d)	26. (a)	27. (d)	28. (c)	29. (d)	30. (c)
31. (d)	32. (d)	33. (c)	34. (d)	35. (a)	36. (c)	37. (b)	38. (b)	39. (c)	40. (c)
41. (b)	42. (b)	43. (c)	44. (d)	45. (b)	46. (c)	47. (a)	48. (a)	49. (c)	50. (c)
51. (b)	52. (b)	53. (a)	54. (a)	55. (c)	56. (b)	57. (c)	58. (a)	59. (c)	60. (a)
61. (c)	62. (a)	63. (a)	64. (c)	65. (a)	66. (c)	67. (a)	68. (d)	69. (b)	70. (d)
71. (c)	72. (c)	73. (c)	74. (a)	75. (c)	76. (c)	77. (b)	78. (a)	79. (b)	80. (c)
81. (b)	82. (a)	83. (a)	84. (a)	85. (b)	86. (b)	87. (d)	88. (b)	89. (d)	90. (b)
91. (b)	92. (c)	93. (c)	94. (b)	95. (d)	96. (b)	97. (b)	98. (d)	99. (a)	100. (b)
101. (d)	102. (b)	103. (c)	104. (b)	105. (c)	106. (d)	107. (d)	108. (d)	109. (c)	110. (a)
111. (a)	112. (b)	113. (c)	114. (c)	115. (b)	116. (c)	117. (d)	118. (d)	119. (b)	120. (d)
121. (d)	122. (b)	123. (c)	124. (b)	125. (d)	126. (d)	127. (c)	128. (b)	129. (b)	130. (b)
131. (c)	132. (b)	133. (b)	134. (b)	135. (c)	136. (d)	137. (b)	138. (c)	139. (a)	140. (b)
141. (b)	142. (d)	143. (a)	144. (c)	145. (a)	146. (b)	147. (b)	148. (b)	149. (c)	150. (c)
151. (b)	152. (c)	153. (a)							

Exercise-2 : One or More than One Answer Is/are Correct

- Let S is the set of all real x such that $\frac{2x-1}{2x^3+3x^2+x}$ is positive, then S contains :
 (a) $\left(-\infty, -\frac{3}{2}\right)$ (b) $\left(-\frac{3}{2}, -\frac{1}{2}\right)$ (c) $\left(-\frac{1}{2}, 0\right)$ (d) $\left(\frac{1}{2}, 2\right)$
- If $kx^2 - 4x + 3k + 1 > 0$ for atleast one $x > 0$, then if $k \in S$, then S contains:
 (a) $(1, \infty)$ (b) $(0, \infty)$ (c) $(-1, \infty)$ (d) $\left(-\frac{1}{4}, \infty\right)$
- The equation $|x^2 - x - 6| = x + 2$ has:
 (a) two positive roots (b) two real roots
 (c) three real roots (d) four real roots
- If the roots of the equation $x^2 - ax - b = 0$ ($a, b \in \mathbb{R}$) are both lying between -2 and 2 , then :
 (a) $|a| < 2 - \frac{b}{2}$ (b) $|a| > 2 - \frac{b}{2}$
 (c) $|a| < 4$ (d) $|a| > \frac{b}{2} - 2$
- Consider the equation in real number x and a real parameter λ , $|x-1| - |x-2| + |x-4| = \lambda$. Then for $\lambda \geq 1$, the number of solutions, the equation can have is/are :
 (a) 1 (b) 2 (c) 3 (d) 4
- If a and b are two distinct non-zero real numbers such that $a - b = \frac{a}{b} = \frac{1}{b} - \frac{1}{a}$, then :
 (a) $a > 0$ (b) $a < 0$ (c) $b < 0$ (d) $b > 0$
- Let $f(x) = ax^2 + bx + c$, $a > 0$ and $f(2-x) = f(2+x) \forall x \in \mathbb{R}$ and $f(x) = 0$ has 2 distinct real roots, then which of the following is true ?
 (a) Atleast one root must be positive
 (b) $f(2) < f(0) > f(1)$
 (c) Minimum value of $f(x)$ is negative
 (d) Vertex of graph of $y = f(x)$ lies in 3rd quadrat
- In the above problem, if roots of equation $f(x) = 0$ are non-real complex, then which of the following is false ?
 (a) $f(x) = \sin \frac{\pi x}{4}$ must have 2 solutions
 (b) $4a - 2b + c < 0$
 (c) If $\log_{f(2)} f(3)$ is not defined, then $f(x) \geq 1 \forall x \in \mathbb{R}$
 (d) All a, b, c are positive
- If exactly two integers lie between the roots of equation $x^2 + ax - 1 = 0$. Then integral value(s) of ' a ' is/are :
 (a) -1 (b) -2 (c) 1 (d) 2

10. If the minimum value of the quadratic expression $y = ax^2 + bx + c$ is negative attained at negative value of x , then :
 (a) $a > 0$ (b) $b > 0$ (c) $c > 0$ (d) $D > 0$
 (where D is discriminant)
11. The quadratic expression $ax^2 + bx + c > 0 \forall x \in R$, then:
 (a) $13a - 5b + 2c > 0$ (b) $13a - b + 2c > 0$
 (c) $c > 0, D < 0$ (d) $a + c > b, D < 0$
 (where D is discriminant)
12. The possible positive integral value of ' k ' for which $5x^2 - 2kx + 1 < 0$ has exactly one integral solution may be divisible by :
 (a) 2 (b) 3 (c) 5 (d) 7
13. If the equation $x^2 + px + q = 0$, the coefficient of x was incorrectly written as 17 instead of 13. Then roots were found to be -2 and -15. The correct roots are :
 (a) -1 (b) -3 (c) -5 (d) -10
14. If $x^2 - 3x + 2 > 0$ and $x^2 - 3x - 4 \leq 0$, then :
 (a) $|x| \leq 2$ (b) $2 \leq x \leq 4$ (c) $-1 \leq x < 1$ (d) $2 < x \leq 4$
15. If $5^x + (2\sqrt{3})^{2x} - 169 \leq 0$ is true for x lying in the interval :
 (a) $(-\infty, 2)$ (b) $(0, 2]$ (c) $(2, \infty)$ (d) $(0, 4)$
16. Let $f(x) = x^2 + ax + b$ and $g(x) = x^2 + cx + d$ be two quadratic polynomials with real coefficients and satisfy $ac = 2(b + d)$. Then which of the following is (are) correct?
 (a) Exactly one of either $f(x) = 0$ or $g(x) = 0$ must have real roots.
 (b) Atleast one of either $f(x) = 0$ or $g(x) = 0$ must have real roots.
 (c) Both $f(x) = 0$ and $g(x) = 0$ must have real roots.
 (d) Both $f(x) = 0$ and $g(x) = 0$ must have imaginary roots.
17. The expression $\frac{1}{\sqrt{x+2\sqrt{x-1}}} + \frac{1}{\sqrt{x-2\sqrt{x-1}}}$ simplifies to :
 (a) $\frac{2}{3-x}$ if $1 < x < 2$ (b) $\frac{2}{2-x}$ if $1 < x < 2$
 (c) $\frac{2\sqrt{x-1}}{(x-2)}$ if $x > 2$ (d) $\frac{2\sqrt{x-1}}{x+2}$ if $x > 2$
18. If all values of x which satisfies the inequality $\log_{(1/3)}(x^2 + 2px + p^2 + 1) \geq 0$ also satisfy the inequality $kx^2 + kx - k^2 \leq 0$ for all real values of k , then all possible values of p lies in the interval :
 (a) $[-1, 1]$ (b) $[0, 1]$ (c) $[0, 2]$ (d) $[-2, 0]$
19. Which of the following statement(s) is/are correct?
 (a) The number of quadratic equations having real roots which remain unchanged even after squaring their roots is 3.

- (b) The number of solutions of the equation $\tan 2\theta + \tan 3\theta = 0$, in the interval $[0, \pi]$ is equal to 6.
- (c) For $x_1, x_2, x_3 > 0$, the minimum value of $\frac{2x_1}{x_2} + \frac{128x_3^2}{x_2^2} + \frac{x_2^3}{4x_1x_3^2}$ equals 24.
- (d) The locus of the mid-points of chords of the circle $x^2 + y^2 - 2x - 6y - 1 = 0$, which are passing through origin is $x^2 + y^2 - x - 3y = 0$.
- 20.** If $(a, 0)$ is a point on a diameter inside the circle $x^2 + y^2 = 4$. Then $x^2 - 4x - a^2 = 0$ has :
- (a) Exactly one real root in $(-1, 0]$ (b) Exactly one real root in $[2, 5]$
 (c) Distinct roots greater than -1 (d) Distinct roots less than 5
- 21.** Let $x^2 - px + q = 0$ where $p \in R, q \in R, pq \neq 0$ have the roots α, β such that $\alpha + 2\beta = 0$, then:
- (a) $2p^2 + q = 0$ (b) $2q^2 + p = 0$ (c) $q < 0$ (d) $q > 0$
- 22.** If a, b, c are rational numbers ($a > b > c > 0$) and quadratic equation $(a + b - 2c)x^2 + (b + c - 2a)x + (c + a - 2b) = 0$ has a root in the interval $(-1, 0)$ then which of the following statement(s) is/are correct ?
- (a) $a + c < 2b$
 (b) both roots are rational
 (c) $ax^2 + 2bx + c = 0$ have both roots negative
 (d) $cx^2 + 2bx + a = 0$ have both roots negative
- 23.** For the quadratic polynomial $f(x) = 4x^2 - 8ax + a$, the statements(s) which hold good is/are:
- (a) There is only one integral 'a' for which $f(x)$ is non-negative $\forall x \in R$
 (b) For $a < 0$, the number zero lies between the zeroes of the polynomial
 (c) $f(x) = 0$ has two distinct solutions in $(0, 1)$ for $a \in \left(\frac{1}{7}, \frac{4}{7}\right)$
 (d) The minimum value of $f(x)$ for minimum value of a for which $f(x)$ is non-negative $\forall x \in R$ is 0
- 24.** Given a, b, c are three distinct real numbers satisfying the inequality $a - 2b + 4c > 0$ and the equation $ax^2 + bx + c = 0$ has no real roots. Then the possible value(s) of $\frac{4a + 2b + c}{a + 3b + 9c}$ is/are :
- (a) 2 (b) -1 (c) 3 (d) $\sqrt{2}$
- 25.** Let $f(x) = x^2 - 4x + c \forall x \in R$, where c is a real constant, then which of the following is/are true ?
- (a) $f(0) > f(1) > f(2)$ (b) $f(2) > f(3) > f(4)$
 (c) $f(1) < f(4) < f(-1)$ (d) $f(0) = f(4) > f(3)$
- 26.** If $0 < a < b < c$ and the roots α, β of the equation $ax^2 + bx + c = 0$ are imaginary, then :
- (a) $|\alpha| = |\beta|$ (b) $|\alpha| > 1$ (c) $|\beta| < 1$ (d) $|\alpha| = 1$
- 27.** If x satisfies $|x - 1| + |x - 2| + |x - 3| > 6$, then :
- (a) $x \in (-\infty, 1)$ (b) $x \in (-\infty, 0)$ (c) $x \in (4, \infty)$ (d) $(2, \infty)$

28. If both roots of the quadratic equation $ax^2 + x + b - a = 0$ are non real and $b > -1$, then which of the following is/are correct?
 (a) $a > 0$ (b) $a < b$ (c) $3a > 2 + 4b$ (d) $3a < 2 + 4b$
29. If a, b are two numbers such that $a^2 + b^2 = 7$ and $a^3 + b^3 = 10$, then :
 (a) The greatest value of $|a + b| = 5$ (b) The greatest value of $(a + b)$ is 4
 (c) The least value of $(a + b)$ is 1 (d) The least value of $|a + b|$ is 1
30. The number of non-negative integral ordered pair(s) (x, y) for which $(xy - 7)^2 = x^2 + y^2$ holds is greater than or equal to :
 (a) 1 (b) 2 (c) 3 (d) 4
31. If α, β, γ and δ are the roots of the equation $x^4 - bx - 3 = 0$; then an equation whose roots are $\frac{\alpha + \beta + \gamma}{\delta^2}, \frac{\alpha + \beta + \delta}{\gamma^2}, \frac{\alpha + \gamma + \delta}{\beta^2}$ and $\frac{\beta + \gamma + \delta}{\alpha^2}$ is :
 (a) $3x^4 + bx + 1 = 0$ (b) $3x^4 - bx + 1 = 0$
 (c) $3x^4 + bx^3 - 1 = 0$ (d) $3x^4 - bx^3 - 1 = 0$
32. The value of k for which both roots of the equation $4x^2 - 2x + k = 0$ are completely in $(-1, 1)$ may be equal to :
 (a) -1 (b) 0 (c) 2 (d) -3
33. If $a, b, c \in R$, then for which of the following graphs of the quadratic polynomial $y = ax^2 - 2bx + c$ ($a \neq 0$); the product (abc) is negative ?

34. If the equation $ax^2 + bx + c = 0$; $a, b, c \in R$ and $a \neq 0$ has no real roots then which of the following is/are always correct ?
 (a) $(a + b + c)(a - b + c) > 0$ (b) $(a + b + c)(a - 2b + 4c) > 0$
 (c) $(a - b + c)(4a - 2b + c) > 0$ (d) $a(b^2 - 4ac) > 0$
35. If α and β are the roots of the equation $ax^2 + bx + c = 0$; $a, b, c \in R$; $a \neq 0$ then which is (are) correct :
 (a) $\alpha^2 + \beta^2 = \frac{b^2 - 2ac}{a^2}$ (b) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{b^2 - 2ac}{c^2}$
 (c) $\frac{1}{\alpha^3} + \frac{1}{\beta^3} = \frac{abc - b^3}{c^3}$ (d) $\alpha\beta(\alpha + \beta) = \frac{-bc}{a^2}$
36. The equation $\cos^2 x - \sin x + \lambda = 0$, $x \in (0, \pi/2)$ has roots then value(s) of λ can be equal to :
 (a) 0 (b) -1 (c) 1/2 (d) 1
37. If the equation $\ln(x^2 + 5x) - \ln(x + a + 3) = 0$ has exactly one solution for x , then possible integral value of a is :
 (a) -3 (b) -1 (c) 0 (d) 2

38. The number of non-negative integral ordered pair(s) (x, y) for which $(xy - 7)^2 = x^2 + y^2$ holds is greater than or equal to :
 (a) 1 (b) 2 (c) 3 (d) 4
39. If $a < 0$, then the value of x satisfying $x^2 - 2a|x - a| - 3a^2 = 0$ is/are
 (a) $a(1 - \sqrt{2})$ (b) $a(1 + \sqrt{2})$ (c) $a(-1 - \sqrt{6})$ (d) $a(-1 + \sqrt{6})$
40. If $0 < a < b < c$ and the roots α, β of the equation $ax^2 + bx + c = 0$ are imaginary, then
 (a) $|\alpha| = |\beta|$ (b) $|\alpha| > 1$ (c) $|\beta| < 1$ (d) $|\alpha| = 1$
41. If x satisfies $|x - 1| + |x - 2| + |x - 3| > 6$, then
 (a) $x \in (-\infty, 1)$ (b) $x \in (-\infty, 0)$ (c) $x \in (4, \infty)$ (d) $(2, \infty)$
42. The value of k for which both roots of the equation $4x^2 - 2x + k = 0$ are completely in $(-1, 1)$, may be equal to :
 (a) -1 (b) 0 (c) 2 (d) -3
43. Let $\alpha, \beta, \gamma, \delta$ are roots of $x^4 - 12x^3 + \lambda x^2 - 54x + 14 = 0$
 If $\alpha + \beta = \gamma + \delta$, then
 (a) $\lambda = 45$ (b) $\lambda = -45$
 (c) If $\alpha^2 + \beta^2 < \gamma^2 + \delta^2$ then $\frac{\alpha\beta}{\gamma\delta} = \frac{7}{2}$ (d) If $\alpha^2 + \beta^2 < \gamma^2 + \delta^2 \Rightarrow \frac{\alpha\beta}{\gamma\delta} = \frac{2}{7}$
44. If $\left(\frac{a^3}{a-1}, \frac{a^2-3}{a-1}\right); \left(\frac{b^3}{b-1}, \frac{b^2-3}{b-1}\right); \left(\frac{c^3}{c-1}, \frac{c^2-3}{c-1}\right)$ lie on $L: lx + my + n = 0$; where a, b, c are real numbers different from 1; then
 (a) $a + b + c = -\frac{m}{l}$ (b) $abc = \frac{m+n}{l}$
 (c) $ab + bc + ca = \frac{n}{l}$ (d) $abc - (ab + bc + ca) + 3(a + b + c) = 0$

Answers

1.	(a, c, d)	2.	(a, b, d)	3.	(a, c)	4.	(a, c, d)	5.	(a, b, c, d)	6.	(b, c)
7.	(a, b, c)	8.	(a, b, d)	9.	(a, c)	10.	(a, b, d)	11.	(a, b, c, d)	12.	(a, c)
13.	(b, d)	14.	(c, d)	15.	(a, b)	16.	(b)	17.	(b, c)	18.	(a, b, c)
19.	(a, b, d)	20.	(a, b, c, d)	21.	(a, c)	22.	(a, b, c, d)	23.	(a, b, d)	24.	(a, c, d)
25.	(a, c, d)	26.	(a, b)	27.	(b, c)	28.	(a, b)	29.	(a, b, d)	30.	(a, b, c, d)
31.	(d)	32.	(a, b)	33.	(a, c, d)	34.	(a, b, d)	35.	(a, b, d)	36.	(a, c)
37.	(b, c, d)	38.	(a, b, c, d)	39.	(a, d)	40.	(a, b)	41.	(b, c)	42.	(a, b)
43.	(a, c)	44.	(a, c, d)								

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

Let $f(x) = ax^2 + bx + c$, $a \neq 0$, such that $f(-1-x) = f(-1+x) \forall x \in R$. Also given that $f(x) = 0$ has no real roots and $4a + b > 0$.

- Let $\alpha = 4a - 2b + c$, $\beta = 9a + 3b + c$, $\gamma = 9a - 3b + c$, then which of the following is correct ?
 (a) $\beta < \alpha < \gamma$ (b) $\gamma < \alpha < \beta$ (c) $\alpha < \gamma < \beta$ (d) $\alpha < \beta < \gamma$
- Let $p = b - 4a$, $q = 2a + b$, then pq is :
 (a) negative (b) positive (c) 0 (d) nothing can be said

Paragraph for Question Nos. 3 to 4

If α, β are the roots of equation $(k+1)x^2 - (20k+14)x + 91k+40 = 0$; $(\alpha < \beta) k > 0$, then answer the following questions.

- The smaller root (α) lie in the interval :
 (a) (4, 7) (b) (7, 10) (c) (10, 13) (d) None of these
- The larger root (β) lie in the interval :
 (a) (4, 7) (b) (7, 10) (c) (10, 13) (d) None of these

Paragraph for Question Nos. 5 to 7

Let $f(x) = x^2 + bx + c \forall x \in R$, ($b, c \in R$) attains its least value at $x = -1$ and the graph of $f(x)$ cuts y-axis at $y = 2$.

- The least value of $f(x) \forall x \in R$ is :
 (a) -1 (b) 0 (c) 1 (d) $3/2$
- The value of $f(-2) + f(0) + f(1) =$
 (a) 3 (b) 5 (c) 7 (d) 9
- If $f(x) = a$ has two distinct real roots, then complete set of values of a is :
 (a) $(1, \infty)$ (b) $(-2, -1)$ (c) $(0, 1)$ (d) $(1, 2)$

Paragraph for Question Nos. 8 to 9

Consider the equation $\log_2^2 x - 4\log_2 x - m^2 - 2m - 13 = 0$, $m \in R$. Let the real roots of the equation be x_1, x_2 such that $x_1 < x_2$.

- The set of all values of m for which the equation has real roots is :
 (a) $(-\infty, 0)$ (b) $(0, \infty)$ (c) $[1, \infty)$ (d) $(-\infty, \infty)$

9. The sum of maximum value of x_1 and minimum value of x_2 is :

- (a) $\frac{513}{8}$ (b) $\frac{513}{4}$ (c) $\frac{1025}{8}$ (d) $\frac{257}{4}$

Paragraph for Question Nos. 10 to 11

The equation $x^4 - 2x^3 - 3x^2 + 4x - 1 = 0$ has four distinct real roots x_1, x_2, x_3, x_4 such that $x_1 < x_2 < x_3 < x_4$ and product of two roots is unity, then :

10. $x_1x_2 + x_1x_3 + x_2x_4 + x_3x_4 =$
 (a) 0 (b) 1 (c) $\sqrt{5}$ (d) -1
 11. $x_2^3 + x_4^3 =$
 (a) $\frac{2+5\sqrt{5}}{8}$ (b) -4 (c) $\frac{27\sqrt{5}+5}{4}$ (d) 18

Paragraph for Question Nos. 12 to 14

Let $f(x)$ be a polynomial of degree 5 with leading coefficient unity, such that $f(1) = 5, f(2) = 4, f(3) = 3, f(4) = 2$ and $f(5) = 1$, then :

12. $f(6)$ is equal to :
 (a) 120 (b) -120 (c) 0 (d) 6
 13. Sum of the roots of $f(x)$ is equal to :
 (a) 15 (b) -15 (c) 21 (d) can't be determine
 14. Product of the roots of $f(x)$ is equal to :
 (a) 120 (b) -120 (c) 114 (d) -114

Paragraph for Question Nos. 15 to 16

Consider the cubic equation in x , $x^3 - x^2 + (x - x^2)\sin\theta + (x - x^2)\cos\theta + (x - 1)\sin\theta\cos\theta = 0$ whose roots are α, β, γ .

15. The value of $\left(\frac{\alpha}{2}\right)^2 + \left(\frac{\beta}{2}\right)^2 + \left(\frac{\gamma}{2}\right)^2 =$
 (a) 1 (b) $\frac{1}{2}$
 (c) $2\cos\theta$ (d) $\frac{1}{2}(\sin\theta + \cos\theta + \sin\theta\cos\theta)$
 16. Number of values of θ in $[0, 2\pi]$ for which at least two roots are equal, is :
 (a) 2 (b) 3 (c) 4 (d) 5

Paragraph for Question Nos. 17 to 18

Let $P(x)$ be a quadratic polynomial with real coefficients such that for all real x the relation $2(1 + P(x)) = P(x-1) + P(x+1)$ holds.

If $P(0) = 8$ and $P(2) = 32$ then :

17. The sum of all the coefficient of $P(x)$ is :
 (a) 20 (b) 19 (c) 17 (d) 15
18. If the range of $P(x)$ is $[m, \infty)$, then the value of m is :
 (a) -12 (b) 15 (c) -17 (d) -5

Paragraph for Question Nos. 19 to 21

Let t be a real number satisfying $2t^3 - 9t^2 + 30 - \lambda = 0$ where $t = x + \frac{1}{x}$ and $\lambda \in \mathbb{R}$.

19. If the above cubic has three real and distinct solutions for x then exhaustive set of value of λ be :
 (a) $3 < \lambda < 10$ (b) $3 < \lambda < 30$ (c) $\lambda = 10$ (d) None of these
20. If the cubic has exactly two real and distinct solutions for x then exhaustive set of values of λ be :
 (a) $\lambda \in (-\infty, 3) \cup (30, \infty)$ (b) $\lambda \in (-\infty, -22) \cup (10, \infty) \cup \{3\}$
 (c) $\lambda \in \{3, 30\}$ (d) None of these
21. If the cubic has four real and distinct solutions for x then exhaustive set of values of λ be :
 (a) $\lambda \in (3, 10)$ (b) $\lambda \in \{3, 10\}$
 (c) $\lambda \in (-\infty, -22) \cup (10, \infty)$ (d) None of these

Paragraph for Question Nos. 22 to 23

Consider a quadratic expression $f(x) = tx^2 - (2t-1)x + (5t-1)$

22. If $f(x)$ can take both positive and negative values then t must lie in the interval
 (a) $\left(\frac{-1}{4}, \frac{1}{4}\right)$ (b) $\left(-\infty, \frac{-1}{4}\right) \cup \left(\frac{1}{4}, \infty\right)$ (c) $\left(\frac{-1}{4}, \frac{1}{4}\right) - \{0\}$ (d) $(-4, 4)$
23. If $f(x)$ is non-negative $\forall x \geq 0$ then t lies in the interval
 (a) $\left[\frac{1}{5}, \frac{1}{4}\right]$ (b) $\left[\frac{1}{4}, \infty\right)$ (c) $\left[\frac{-1}{4}, \frac{1}{4}\right]$ (d) $\left[\frac{1}{5}, \infty\right)$

Answers

1. (c)	2. (a)	3. (a)	4. (c)	5. (c)	6. (d)	7. (a)	8. (d)	9. (d)	10. (b)
11. (d)	12. (a)	13. (a)	14. (c)	15. (b)	16. (d)	17. (b)	18. (c)	19. (c)	20. (b)
21. (a)	22. (c)	23. (d)							

Exercise-4 : Matching Type Problems

1.

(A)	The least positive integer x , for which $\frac{2x-1}{2x^3+3x^2+x}$ is positive, is equal to	(P)	$\frac{4}{3}$
(B)	If the quadratic equation $3x^2 + 2(a^2 + 1)x + (a^2 - 3a + 2) = 0$ possess roots of opposite sign then a can be equal to	(Q)	1
(C)	The roots of the equation $\sqrt{x+3} - 4\sqrt{x-1} + \sqrt{x+8} - 6\sqrt{x-1} = 1$ can be equal to	(R)	6
(D)	If the roots of the equation $x^4 - 8x^3 + bx^2 - cx + 16 = 0$ are all real and positive then $2(c - b)$ is equal to	(S)	16
		(T)	10

 2. Given the inequality $ax + k^2 > 0$. The complete set of values of ' a ' so that

(A)	The inequality is valid for all values of x and k is	(P)	R
(B)	There exists a value of x such that the inequality is valid for any value of k is	(Q)	ϕ
(C)	There exists a value of k such that the inequality is valid for all values of x is	(R)	$\{0\}$
(D)	There exists values of x and k for which inequality is valid is	(S)	$R - \{0\}$
		(T)	$\{1\}$

3.

(A)	The real root(s) of the equation $x^4 - 8x^2 - 9 = 0$ are	(P)	No real roots
(B)	The real root(s) of the equation $x^{2/3} + x^{1/3} - 2 = 0$ are	(Q)	-3, 3
(C)	The real root(s) of the equation $\sqrt{3x+1} + 1 = \sqrt{x}$ are	(R)	-8, 1

(D)	The real root(s) of the equation $9^x - 10(3^x) + 9 = 0$ are	(S)	0, 2
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4.

(A)	If a, b are the roots of equation $x^2 + ax + b = 0$ ($a, b \in R$), then the number of ordered pairs (a, b) is equal to	(P)	1
(B)	If $P = \operatorname{cosec} \frac{\pi}{8} + \operatorname{cosec} \frac{2\pi}{8} + \operatorname{cosec} \frac{3\pi}{8} + \operatorname{cosec} \frac{13\pi}{8} + \operatorname{cosec} \frac{14\pi}{8} + \operatorname{cosec} \frac{15\pi}{8}$ and $Q = 8 \sin \frac{\pi}{18} \sin \frac{5\pi}{18} \sin \frac{7\pi}{18}$, then $P + Q$ is equal to	(Q)	2
(C)	Let $a_1, a_2, a_3, \dots$ be positive terms of a G.P and $a_4, 1, 2, a_{10}$ are the consecutive terms of another G.P. If $\prod_{i=2}^{12} a_i = 4^{\frac{m}{n}}$ where m and n are coprime, then $(m + n)$ equals	(R)	3
(D)	For $x, y \in R$, if $x^2 - 2xy + 2y^2 - 6y + 9 = 0$, then the value of $5x - 4y$ is equal to	(S)	15

Answers

1.	$A \rightarrow Q; B \rightarrow P; C \rightarrow R; D \rightarrow S$
2.	$A \rightarrow Q; B \rightarrow S; C \rightarrow R; D \rightarrow P$
3.	$A \rightarrow Q; B \rightarrow R; C \rightarrow P; D \rightarrow S$
4.	$A \rightarrow Q; B \rightarrow P; C \rightarrow S; D \rightarrow R$

Exercise-5 : Subjective Type Problems

- Let $f(x) = ax^2 + bx + c$ where a, b, c are integers. If $\sin \frac{\pi}{7} \cdot \sin \frac{3\pi}{7} + \sin \frac{3\pi}{7} \cdot \sin \frac{5\pi}{7} + \sin \frac{5\pi}{7} \cdot \sin \frac{\pi}{7} = f\left(\cos \frac{\pi}{7}\right)$, then find the value of $f(2)$:
- Let a, b, c, d be distinct integers such that the equation $(x-a)(x-b)(x-c)(x-d) - 9 = 0$ has an integer root ' r ', then the value of $a + b + c + d - 4r$ is equal to :
- Consider the equation $(x^2 + x + 1)^2 - (m-3)(x^2 + x + 1) + m = 0$, where m is a real parameter. The number of positive integral values of m for which equation has two distinct real roots, is :
- The number of positive integral values of m , $m \leq 16$ for which the equation given in the above questions has 4 distinct real root is :
- If the equation $(m^2 - 12)x^4 - 8x^2 - 4 = 0$ has no real roots, then the largest value of m is $p\sqrt{q}$ where p, q are coprime natural numbers, then $p + q =$
- The least positive integral value of ' x ' satisfying $(e^x - 2) \left(\sin \left(x + \frac{\pi}{4} \right) \right) (x - \log_e 2) (\sin x - \cos x) < 0$ is :
- The integral values of x for which $x^2 + 17x + 71$ is perfect square of a rational number are a and b , then $|a - b| =$
- Let $P(x) = x^6 - x^5 - x^3 - x^2 - x$ and $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 - x^3 - x^2 - 1 = 0$, then $P(\alpha) + P(\beta) + P(\gamma) + P(\delta) =$
- The number of real values of ' a ' for which the largest value of the function $f(x) = x^2 + ax + 2$ in the interval $[-2, 4]$ is 6 will be :
- The number of all values of n , (where n is a whole number) for which the equation $\frac{x-8}{n-10} = \frac{n}{x}$ has no solution.
- The number of negative integral values of m for which the expression $x^2 + 2(m-1)x + m + 5$ is positive $\forall x > 1$ is :
- If the expression $ax^4 + bx^3 - x^2 + 2x + 3$ has the remainder $4x + 3$ when divided by $x^2 + x - 2$, then $a + 4b = \dots$
- Find the smallest value of k for which both the roots of equation $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values atleast 4.
- If $x^2 - 3x + 2$ is a factor of $x^4 - px^2 + q = 0$, then $p + q =$
- The sum of all real values of k for which the expression $x^2 + 2xy + ky^2 + 2x + k = 0$ can be resolved into linear factors is :
- The curve $y = (a+1)x^2 + 2$ meets the curve $y = ax + 3$, $a \neq -1$ in exactly one point, then $a^2 =$

17. Find the number of integral values of 'a' for which the range of function $f(x) = \frac{x^2 - ax + 1}{x^2 - 3x + 2}$ is $(-\infty, \infty)$.
18. When x^{100} is divided by $x^2 - 3x + 2$, the remainder is $(2^{k+1} - 1)x - 2(2^k - 1)$, then $k =$
19. Let $P(x)$ be a polynomial equation of least possible degree, with rational coefficients, having $\sqrt[3]{7} + \sqrt[3]{49}$ as one of its roots. Then the product of all the roots of $P(x) = 0$ is :
20. The range of values k for which the equation $2\cos^4 x - \sin^4 x + k = 0$ has atleast one solution is $[\lambda, \mu]$. Find the value of $(9\mu + \delta)$.
21. Let $P(x)$ be a polynomial with real coefficient and $P(x) - P'(x) = x^2 + 2x + 1$. Find $P(1)$.
22. Find the smallest positive integral value of a for which the greater root, of the equation $x^2 - (a^2 + a + 1)x + a(a^2 + 1) = 0$ lies between the roots of the equation $x^2 - a^2x - 2(a^2 - 2) = 0$
23. If the equation $x^4 + kx^2 + k = 0$ has exactly two distinct real roots, then the smallest integral value of $|k|$ is :
24. Let a, b, c, d be the roots of $x^4 - x^3 - x^2 - 1 = 0$. Also consider $P(x) = x^6 - x^5 - x^3 - x^2 - x$, then the value of $P(a) + P(b) + P(c) + P(d)$ is equal to :
25. The number of integral values of $a, a \in [-5, 5]$ for which the equation $x^2 + 2(a-1)x + a + 5 = 0$ has one root smaller than 1 and the other root greater than 3 is :
26. The number of non-negative integral values of $n, n \leq 10$ so that a root of the equation $n^2 \sin^2 x - 2 \sin x - (2n + 1) = 0$ lies in interval $\left[0, \frac{\pi}{2}\right]$ is :
27. Let $f(x) = ax^2 + bx + c$, where a, b, c are integers and $a > 1$. If $f(x)$ takes the value p , a prime for two distinct integer values of x , then the number of integer values of x for which $f(x)$ takes the value $2p$ is :
28. If x and y are real numbers connected by the equation $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, then the sum of maximum value of x and the minimum value of y is :
29. Consider two numbers a, b , sum of which is 3 and the sum of their cubes is 7. Then sum of all possible distinct values of a is :
30. If $y^2(y^2 - 6) + x^2 - 8x + 24 = 0$ and the minimum value of $x^2 + y^4$ is m and maximum value is M ; then find the value of $M - 2m$.
31. Consider the equation $x^3 - ax^2 + bx - c = 0$, where a, b, c are rational number, $a \neq 1$. It is given that x_1, x_2 and $x_1 x_2$ are the real roots of the equation. If $(b + c) = 2(a + 1)$, then $x_1 x_2 \left(\frac{a+1}{b+c} \right) =$
32. Let α satisfy the equation $x^3 + 3x^2 + 4x + 5 = 0$ and β satisfy the equation $x^3 - 3x^2 + 4x - 5 = 0, \alpha, \beta \in \mathbb{R}$, then $\alpha + \beta =$

33. Let x, y and z are positive reals and $x^2 + xy + y^2 = 2; y^2 + yz + z^2 = 1$ and $z^2 + zx + x^2 = 3$.
If the value of $xy + yz + zx$ can be expressed as $\sqrt{\frac{p}{q}}$ where p and q are relatively prime positive integer find the value of $p - q$:
34. The number of ordered pairs (a, b) , where a, b are integers satisfying the inequality $\min(x^2 + (a - b)x + (1 - a - b)) > \max(-x^2 + (a + b)x - (1 + a + b)) \forall x \in R$, is :
35. The real value of x satisfying $\sqrt[3]{20x + \sqrt{20x + 13}} = 13$ can be expressed as $\frac{a}{b}$ where a and b are relatively prime positive integers. Find the value of b ?
36. If the range of the values of a for which the roots of the equation $x^2 - 2x - a^2 + 1 = 0$ lie between the roots of the equation $x^2 - 2(a + 1)x + a(a - 1) = 0$ is (p, q) , then find the value of $\left(q - \frac{1}{p}\right)$.
37. Find the number of positive integers satisfying the inequality $x^2 - 10x + 16 < 0$.
38. If $\sin \theta$ and $\cos \theta$ are the roots of the quadratic equation $ax^2 + bx + c = 0$ ($ac \neq 0$). Then find the value of $\frac{b^2 - a^2}{ac}$.
39. Let the inequality $\sin^2 x + a \cos x + a^2 \geq 1 + \cos x$ is satisfied $\forall x \in R$, for $a \in (-\infty, k_1] \cup [k_2, \infty)$, then $|k_1| + |k_2| =$
40. α and β are roots of the equation $2x^2 - 35x + 2 = 0$. Find the value of $\sqrt{(2\alpha - 35)^3 (2\beta - 35)^3}$
41. The sum of all integral values of ' a ' for which the equation $2x^2 - (1 + 2a)x + 1 + a = 0$ has a integral root.
42. Let $f(x)$ be a polynomial of degree 8 such that $F(r) = \frac{1}{r}, r = 1, 2, 3, \dots, 8, 9$, then $\frac{1}{F(10)} =$
43. Let α, β are two real roots of equation $x^2 + px + q = 0, p, q \in R, q \neq 0$. If the quadratic equation $g(x) = 0$ has two roots $\alpha + \frac{1}{\alpha}, \beta + \frac{1}{\beta}$ such that sum of its roots is equal to product of roots, then then number of integral values q can attain is :
44. If $\cos A, \cos B$ and $\cos C$ are the roots of cubic $x^3 + ax^2 + bx + c = 0$, where A, B, C are the angles of a triangle then find the value of $a^2 - 2b - 2c$.
45. Find the number of positive integral values of k for which $kx^2 + (k - 3)x + 1 < 0$ for atleast one positive x .

Answers

1. 9	2. 0	3. 1	4. 7	5. 5	6. 3	7. 3	8. 6	9. 0	10. 6
11. 0	12. 9	13. 2	14. 9	15. 2	16. 4	17. 0	18. 99	19. 56	20. 7
21. 2	22. 3	23. 1	24. 6	25. 4	26. 8	27. 0	28. 7	29. 3	30. 4
31. 1	32. 0	33. 5	34. 9	35. 5	36. 5	37. 5	38. 2	39. 3	40. 8
41. 1	42. 5	43. 3	44. 1	45. 0					

□□□

SEQUENCE AND SERIES

Exercise-1 : Single Choice Problems

- If a, b, c are positive numbers and $a + b + c = 1$, then the maximum value of $(1-a)(1-b)(1-c)$ is :
 (a) 1 (b) $\frac{2}{3}$ (c) $\frac{8}{27}$ (d) $\frac{4}{9}$
- If $xyz = (1-x)(1-y)(1-z)$ where $0 \leq x, y, z \leq 1$, then the minimum value of $x(1-z) + y(1-x) + z(1-y)$ is :
 (a) $\frac{3}{2}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{1}{2}$
- If $\sec(\alpha - 2\beta), \sec \alpha, \sec(\alpha + 2\beta)$ are in arithmetical progression then $\cos^2 \alpha = \lambda \cos^2 \beta$ ($\beta \neq n\pi, n \in I$) the value of λ is :
 (a) 1 (b) 2 (c) 3 (d) $\frac{1}{2}$
- Let a, b, c, d, e are non-zero and distinct positive real numbers. If a, b, c are in A.P ; b, c, d are in G.P and c, d, e are in H.P, then a, c, e are in :
 (a) A.P (b) G.P
 (c) H.P (d) Nothing can be said
- If $(m+1)^{\text{th}}, (n+1)^{\text{th}},$ and $(r+1)^{\text{th}}$ terms of a non-constant A.P are in G.P and m, n, r are in H.P, then the ratio of first term of the A.P to its common difference is :
 (a) $-\frac{n}{2}$ (b) $-n$ (c) $-2n$ (d) $+n$
- If the equation $x^4 - 4x^3 + ax^2 + bx + 1 = 0$ has four positive roots, then the value of $(a+b)$ is :
 (a) -4 (b) 2
 (c) 6 (d) can not be determined

7. If S_1, S_2 and S_3 are the sums of first n natural numbers, their squares and their cubes respectively, then $\frac{S_1^4 S_2^2 - S_2^2 S_3^2}{S_1^2 + S_3^2} =$
 (a) 4 (b) 2 (c) 1 (d) 0
8. If $S_n = \frac{1 \cdot 2}{3!} + \frac{2 \cdot 2^2}{4!} + \frac{3 \cdot 2^3}{5!} + \dots$ upto n terms then the sum of the infinite terms is :
 (a) 1 (b) $\frac{2}{3}$ (c) e (d) $\frac{\pi}{4}$
9. If $\tan\left(\frac{\pi}{12} - x\right), \tan\frac{\pi}{12}, \tan\left(\frac{\pi}{12} + x\right)$ in order are three consecutive terms of a G.P then sum of all the solutions in $[0, 314]$ is $k\pi$. The value of k is :
 (a) 4950 (b) 5050 (c) 2525 (d) 5010
10. Let $S_k = 1 + 2 + 3 + \dots + k$ and $Q_n = \frac{S_2}{S_2 - 1} \cdot \frac{S_3}{S_3 - 1} \cdot \frac{S_4}{S_4 - 1} \dots \frac{S_n}{S_n - 1}$, where $k, n \in N$
 $\lim_{n \rightarrow \infty} Q_n =$
 (a) $\frac{1}{3}$ (b) 1 (c) 3 (d) 0
11. l, m, n are the $p^{\text{th}}, q^{\text{th}}$ and r^{th} term of a G.P all positive, then $\begin{vmatrix} \log l & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix}$ equals :
 (a) -1 (b) 2 (c) 1 (d) 0
12. The number of natural numbers < 300 that are divisible by 6 but not by 9 is :
 (a) 49 (b) 37 (c) 33 (d) 16
13. If $x, y, z > 0$ and $x + y + z = 1$ then $\frac{xyz}{(1-x)(1-y)(1-z)}$ is necessarily.
 (a) ≥ 8 (b) $\leq \frac{1}{8}$ (c) 1 (d) None of these
14. If the roots of the equation $px^2 + qx + r = 0$, where $2p, q, 2r$ are in G.P, are of the form $\alpha^2, 4\alpha - 4$. Then the value of $2p + 4q + 7r$ is :
 (a) 0 (b) 10 (c) 14 (d) 18
15. Let $x_1, x_2, x_3, \dots, x_k$ be the divisors of positive integer n (including 1 and n). If $x_1 + x_2 + x_3 + \dots + x_k = 75$. Then $\sum_{i=1}^k \left(\frac{1}{x_i}\right)$ is equal to :
 (a) $\frac{75}{k}$ (b) $\frac{75}{n}$ (c) $\frac{1}{n}$ (d) $\frac{1}{75}$

16. If $a_1, a_2, a_3, \dots, a_n$ are in H.P. and $f(k) = \sum_{r=1}^n a_r - a_k$ then $\frac{a_1}{f(1)}, \frac{a_2}{f(2)}, \frac{a_3}{f(3)}, \dots, \frac{a_n}{f(n)}$ are in:
- (a) A.P. (b) G.P. (c) H.P. (d) None of these
17. If α, β be roots of the equation $375x^2 - 25x - 2 = 0$ and $s_n = \alpha^n + \beta^n$, then $\lim_{n \rightarrow \infty} \left(\sum_{r=1}^n S_r \right) = \dots$
- (a) $\frac{1}{12}$ (b) $\frac{1}{4}$ (c) $\frac{1}{3}$ (d) 1
18. If $a_i, i = 1, 2, 3, 4$ be four real members of the same sign, then the minimum value of $\sum \frac{a_i}{a_j}, i, j \in \{1, 2, 3, 4\}, i \neq j$ is :
- (a) 6 (b) 8 (c) 12 (d) 24
19. Given that x, y, z are positive reals such that $xyz = 32$. The minimum value of $x^2 + 4xy + 4y^2 + 2z^2$ is equal to :
- (a) 64 (b) 256 (c) 96 (d) 216
20. In an A.P., five times the fifth term is equal to eight times the eighth term. Then the sum of the first twenty five terms is equal to :
- (a) 25 (b) $\frac{25}{2}$ (c) -25 (d) 0
21. Let α, β be two distinct values of x lying in $[0, \pi]$ for which $\sqrt{5} \sin x, 10 \sin x, 10(4 \sin^2 x + 1)$ are 3 consecutive terms of a G.P. Then minimum value of $|\alpha - \beta| =$
- (a) $\frac{\pi}{10}$ (b) $\frac{\pi}{5}$ (c) $\frac{2\pi}{5}$ (d) $\frac{3\pi}{5}$
22. In an infinite G.P., the sum of first three terms is 70. If the extreme terms are multiplied by 4 and the middle term is multiplied by 5, the resulting terms form an A.P. then the sum to infinite terms of G.P. is :
- (a) 120 (b) 40 (c) 160 (d) 80
23. The value of the sum $\sum_{k=1}^{\infty} \sum_{n=1}^{\infty} \frac{k}{2^{n+k}}$ is equal to :
- (a) 5 (b) 4 (c) 3 (d) 2
24. Let p, q, r are positive real numbers, such that $27pqr \geq (p+q+r)^3$ and $3p+4q+5r=12$, then $p^3+q^4+r^5=$
- (a) 3 (b) 6 (c) 2 (d) 4
25. Find the sum of the infinite series $\frac{1}{9} + \frac{1}{18} + \frac{1}{30} + \frac{1}{45} + \frac{1}{63} + \dots$
- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{1}{5}$ (d) $\frac{2}{3}$

26. If S_r denote the sum of first 'r' terms of a non constant A.P and $\frac{S_a}{a^2} = \frac{S_b}{b^2} = c$, where a, b, c are distinct then $S_c =$
 (a) c^2 (b) c^3 (c) c^4 (d) abc
27. In an infinite G.P second term is x and its sum is 4, then complete set of values of ' x ' is :
 (a) $(-8, 0)$ (b) $\left[-\frac{1}{8}, \frac{1}{8}\right] - \{0\}$
 (c) $\left[-1, -\frac{1}{8}\right) \cup \left(\frac{1}{8}, 1\right]$ (d) $(-8, 1] - \{0\}$
28. The number of terms of an A.P is odd. The sum of the odd terms ($1^{\text{st}}, 3^{\text{rd}}$ etc.,) is 248 and the sum of the even terms is 217. The last term exceeds the first by 56, then :
 (a) the number of terms is 17 (b) the first term is 3
 (c) the number of terms is 13 (d) the first term is 1
29. Let $A_1, A_2, A_3, \dots, A_n$ be squares such that for each $n \geq 1$ the length of a side of A_n equals the length of a diagonal of A_{n+1} . If the side of A_1 be 20 units then the smallest value of ' n ' for which area of A_n is less than 1.
 (a) 7 (b) 8 (c) 9 (d) 10
30. Let $S_k = \sum_{i=0}^{\infty} \frac{1}{(k+1)^i}$, then $\sum_{k=1}^n kS_k$ equal :
 (a) $\frac{n(n+1)}{2}$ (b) $\frac{n(n-1)}{2}$ (c) $\frac{n(n+2)}{2}$ (d) $\frac{n(n+3)}{2}$
31. The sum of the series $\frac{2}{1 \cdot 2} + \frac{5}{2 \cdot 3} 2^1 + \frac{10}{3 \cdot 4} 2^2 + \frac{17}{4 \cdot 5} 2^3 + \dots$ upto n terms is equal :
 (a) $\frac{n2^n}{n+1}$ (b) $\left(\frac{n}{n+1}\right) 2^n + 1$ (c) $\frac{n2^n}{n+1} - 1$ (d) $\frac{(n-1)2^n}{n+1}$
32. If $(1 \cdot 5)^{30} = k$, then the value of $\sum_{n=2}^{29} (1 \cdot 5)^n$, is :
 (a) $2k - 3$ (b) $k + 1$ (c) $2k + 7$ (d) $2k - \frac{9}{2}$
33. n arithmetic means are inserted between 7 and 49 and their sum is found to be 364, then n is :
 (a) 11 (b) 12 (c) 13 (d) 14
34. The third term of a G.P is 2. Then the product of the first five terms, is :
 (a) 2^3 (b) 2^4 (c) 2^5 (d) none of these
35. The sum of first n terms of an A.P is $5n^2 + 4n$, its common difference is :
 (a) 9 (b) 10 (c) 3 (d) -4

36. If $x + y = a$ and $x^2 + y^2 = b$, then the value of $(x^3 + y^3)$, is :
 (a) ab (b) $a^2 + b$ (c) $a + b^2$ (d) $\frac{3ab - a^3}{2}$
37. If $S_1, S_2, S_3, \dots, S_n$ are the sum of infinite geometric series whose first terms are $1, 3, 5, \dots, (2n-1)$ and whose common ratios are $\frac{2}{3}, \frac{2}{5}, \dots, \frac{2}{2n+1}$ respectively, then

$$\left\{ \frac{1}{S_1 S_2 S_3} + \frac{1}{S_2 S_3 S_4} + \frac{1}{S_3 S_4 S_5} + \dots \text{upto infinite terms} \right\} =$$

 (a) $\frac{1}{15}$ (b) $\frac{1}{60}$ (c) $\frac{1}{12}$ (d) $\frac{1}{3}$
38. Sequence $\{t_n\}$ of positive terms is a G.P. If $t_6, 2, 5, t_{14}$ form another G.P. in that order, then the product $t_1 t_2 t_3 \dots t_{18} t_{19}$ is equal to :
 (a) 10^9 (b) 10^{10} (c) $10^{17/2}$ (d) $10^{19/2}$
39. The minimum value of $\frac{(A^2 + A + 1)(B^2 + B + 1)(C^2 + C + 1)(D^2 + D + 1)}{ABCD}$ where $A, B, C, D > 0$ is :
 (a) $\frac{1}{3^4}$ (b) $\frac{1}{2^4}$ (c) 2^4 (d) 3^4
40. If $\sum_{r=1}^{20} r^3 = a$, $\sum_{r=1}^{20} r^2 = b$ then sum of products of 1, 2, 3, 4, 20 taking two at a time is :
 (a) $\frac{a-b}{2}$ (b) $\frac{a^2 - b^2}{2}$ (c) $a - b$ (d) $a^2 - b^2$
41. The sum of the first $2n$ terms of an A.P. is x and the sum of the next n terms is y , its common difference is :
 (a) $\frac{x-2y}{3n^2}$ (b) $\frac{2y-x}{3n^2}$ (c) $\frac{x-2y}{3n}$ (d) $\frac{2y-x}{3n}$
42. The number of non-negative integers 'n' satisfying $n^2 = p + q$ and $n^3 = p^2 + q^2$ where p and q are integers.
 (a) 2 (b) 3 (c) 4 (d) Infinite
43. Concentric circles of radii 1, 2, 3 100 cms are drawn. The interior of the smallest circle is coloured red and the angular regions are coloured alternately green and red, so that no two adjacent regions are of the same colour. The total area of the green regions in sq. cm is equals to:
 (a) 1000π (b) 5050π (c) 4950π (d) 5151π
44. If $\log_2 4, \log_{\sqrt{2}} 8$ and $\log_3 9^{k-1}$ are consecutive terms of a geometric sequence, then the number of integers that satisfy the system of inequalities $x^2 - x > 6$ and $|x| < k^2$ is :
 (a) 193 (b) 194 (c) 195 (d) 196

45. Let T_r be the r^{th} term of an A.P whose first term is $-\frac{1}{2}$ and common difference is 1, then

$$\sum_{r=1}^n \sqrt{1 + T_r T_{r+1} T_{r+2} T_{r+3}} =$$

- (a) $\frac{n(n+1)(2n+1)}{6} - \frac{5n}{4}$ (b) $\frac{n(n+1)(2n+1)}{6} - \frac{5n}{4} + \frac{1}{4}$
 (c) $\frac{n(n+1)(2n+1)}{6} - \frac{5n}{4} + \frac{1}{2}$ (d) $\frac{n(n+1)(2n+1)}{12} - \frac{5n}{8} + 1$

46. If $\sum_{r=1}^n T_r = \frac{n(n+1)(n+2)}{3}$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{2008}{T_r} =$

- (a) 2008 (b) 3012 (c) 4016 (d) 8032

47. The sum of the infinite series, $1^2 - \frac{2^2}{5} + \frac{3^2}{5^2} - \frac{4^2}{5^3} + \frac{5^2}{5^4} - \frac{6^2}{5^5} + \dots$ is :

- (a) $\frac{1}{2}$ (b) $\frac{25}{24}$ (c) $\frac{25}{54}$ (d) $\frac{125}{252}$

48. The absolute term in $P(x) = \sum_{r=1}^n \left(x - \frac{1}{r}\right) \left(x - \frac{1}{r+1}\right) \left(x - \frac{1}{r+2}\right)$ as n approaches to infinity is :

- (a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $-\frac{1}{4}$

49. Let a, b, c are positive real numbers such that $p = a^2b + ab^2 - a^2c - ac^2$; $q = b^2c + bc^2 - a^2b - ab^2$ and $r = ac^2 + a^2c - cb^2 - bc^2$ and the quadratic equation $px^2 + qx + r = 0$ has equal roots ; then a, b, c are in :

- (a) A.P (b) G.P (c) H.P (d) None of these

50. If T_k denotes the k^{th} term of an H.P from the beginning and $\frac{T_2}{T_6} = 9$, then $\frac{T_{10}}{T_4}$ equals :

- (a) $\frac{17}{5}$ (b) $\frac{5}{17}$ (c) $\frac{7}{19}$ (d) $\frac{19}{7}$

51. Number of terms common to the two sequences 17, 21, 25,, 417 and 16, 21, 26,, 466 is :

- (a) 19 (b) 20 (c) 21 (d) 22

52. The sum of the series $1 + \frac{2}{3} + \frac{1}{3^2} + \frac{2}{3^3} + \frac{1}{3^4} + \frac{2}{3^5} + \frac{1}{3^6} + \frac{2}{3^7} + \dots$ upto infinite terms is equal to :

- (a) $\frac{15}{8}$ (b) $\frac{8}{15}$ (c) $\frac{27}{8}$ (d) $\frac{21}{8}$

53. The coefficient of x^8 in the polynomial $(x-1)(x-2)(x-3)\dots(x-10)$ is :

- (a) 2640 (b) 1320 (c) 1370 (d) 2740

54. Let $\alpha = \lim_{n \rightarrow \infty} \frac{(1^3 - 1^2) + (2^3 - 2^2) + \dots + (n^3 - n^2)}{n^4}$, then α is equal to :
- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) non-existent
55. If $16x^4 - 32x^3 + ax^2 + bx + 1 = 0$, $a, b \in R$ has positive real roots only, then $a - b$ is equal to :
- (a) -32 (b) 32 (c) 49 (d) -49
56. If ABC is a triangle and $\tan \frac{A}{2}, \tan \frac{B}{2}, \tan \frac{C}{2}$ are in H.P, then the minimum value of $\cot \frac{B}{2} =$
- (a) $\sqrt{3}$ (b) 1 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{\sqrt{3}}$
57. If α and β are the roots of the quadratic equation $4x^2 + 2x - 1 = 0$ then the value of $\sum_{r=1}^{\infty} (\alpha^r + \beta^r)$ is :
- (a) 2 (b) 3 (c) 6 (d) 0
58. The sum of the series $2^2 + 2(4)^2 + 3(6)^2 + \dots$ upto 10 terms is equal to :
- (a) 11300 (b) 12100 (c) 12300 (d) 11200
59. If a and b are positive real numbers such that $a + b = 6$, then the minimum value of $\left(\frac{4}{a} + \frac{1}{b}\right)$ is equal to :
- (a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) 1 (d) $\frac{3}{2}$
60. The first term of an infinite G.P is the value of x satisfying the equation $\log_4(4^x - 15) + x - 2 = 0$ and the common ratio is $\cos\left(\frac{2011\pi}{3}\right)$. The sum of G.P is :
- (a) 1 (b) $\frac{4}{3}$ (c) 4 (d) 2
61. Let a, b, c be positive numbers, then the minimum value of $\frac{a^4 + b^4 + c^2}{abc}$ is :
- (a) 4 (b) $2^{3/4}$ (c) $\sqrt{2}$ (d) $2\sqrt{2}$
62. If $xy = 1$; then minimum value of $x^2 + y^2$ is :
- (a) 1 (b) 2 (c) $\sqrt{2}$ (d) 4
63. Find the value of $\frac{2}{1^3} + \frac{6}{1^3 + 2^3} + \frac{12}{1^3 + 2^3 + 3^3} + \frac{20}{1^3 + 2^3 + 3^3 + 4^3} + \dots$ upto 60 terms :
- (a) 2 (b) $\frac{1}{2}$ (c) 4 (d) $\frac{1}{4}$

64. Evaluate : $\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)(n+3)\dots(n+k)}$
- (a) $\frac{1}{(k-1)(k-1)!}$ (b) $\frac{1}{k \cdot k!}$ (c) $\frac{1}{(k-1)k!}$ (d) $\frac{1}{k!}$
65. Consider two positive numbers a and b . If arithmetic mean of a and b exceeds their geometric mean by $3/2$ and geometric mean of a and b exceeds their harmonic mean by $6/5$ then the value of $a^2 + b^2$ will be :
- (a) 150 (b) 153 (c) 156 (d) 159
66. Sum of first 10 terms of the series, $S = \frac{7}{2^2 \cdot 5^2} + \frac{13}{5^2 \cdot 8^2} + \frac{19}{8^2 \cdot 11^2} + \dots$ is :
- (a) $\frac{255}{1024}$ (b) $\frac{88}{1024}$ (c) $\frac{264}{1024}$ (d) $\frac{85}{1024}$
67. $\sum_{r=1}^{10} \frac{r}{1-3r^2+r^4} =$
- (a) $-\frac{50}{109}$ (b) $-\frac{54}{109}$ (c) $-\frac{55}{111}$ (d) $-\frac{55}{109}$
68. Let r^{th} term t_r of a series is given by $t_r = \frac{r}{1+r^2+r^4}$. Then $\lim_{n \rightarrow \infty} \sum_{r=1}^n t_r$ is equal to :
- (a) $\frac{1}{2}$ (b) 1 (c) 2 (d) $\frac{1}{4}$
69. The sum of the series $1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \dots$ to infinite terms, is :
- (a) $\frac{31}{12}$ (b) $\frac{41}{16}$ (c) $\frac{45}{16}$ (d) $\frac{35}{16}$
70. The third term of a G.P is 2. Then the product of the first five terms, is :
- (a) 2^3 (b) 2^4 (c) 2^5 (d) none of these
71. If $x_1, x_2, x_3, \dots, x_{2n}$ are in A.P, then $\sum_{r=1}^{2n} (-1)^{r+1} x_r^2$ is equal to :
- (a) $\frac{n}{(2n-1)} (x_1^2 - x_{2n}^2)$ (b) $\frac{2n}{(2n-1)} (x_1^2 - x_{2n}^2)$
- (c) $\frac{n}{n-1} (x_1^2 - x_{2n}^2)$ (d) $\frac{n}{2n+1} (x_1^2 - x_{2n}^2)$
72. Let two numbers have arithmetic mean 9 and geometric mean 4. Then these numbers are roots of the equation :
- (a) $x^2 + 18x + 16 = 0$ (b) $x^2 - 18x - 16 = 0$
- (c) $x^2 + 18x - 16 = 0$ (d) $x^2 - 18x + 16 = 0$

73. If p and q are positive real numbers such that $p^2 + q^2 = 1$, then the maximum value of $(p + q)$ is :
 (a) 2 (b) $\frac{1}{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$
74. A person has to count 4500 currency notes. Let a_n denote the number of notes he counts in the n^{th} minute. If $a_1 = a_2 = \dots = a_{10} = 150$ and $a_{10}, a_{11}, a_{12}, \dots$ are in A.P. with common difference -2 , then the time taken by him to count all notes is :
 (a) 34 minutes (b) 24 minutes (c) 125 minutes (d) 35 minutes
75. A non constant arithmetic progression has common difference d and first term is $(1 - ad)$. If the sum of the first 20 terms is 20, then the value of a is equal to :
 (a) $\frac{2}{19}$ (b) $\frac{19}{2}$ (c) $\frac{2}{9}$ (d) $\frac{9}{2}$
76. The value of $\sum_{n=3}^{\infty} \frac{1}{n^5 - 5n^3 + 4n} =$
 (a) $\frac{1}{120}$ (b) $\frac{1}{96}$ (c) $\frac{1}{24}$ (d) $\frac{1}{144}$
77. Find the value of $\frac{2}{1^3} + \frac{6}{1^3 + 2^3} + \frac{12}{1^3 + 2^3 + 3^3} + \frac{20}{1^3 + 2^3 + 3^3 + 4^3} + \dots$ up to infinite terms:
 (a) 2 (b) $\frac{1}{2}$ (c) 4 (d) $\frac{1}{4}$
78. The minimum value of the expression $2^x + 2^{2x+1} + \frac{5}{2^x}$, $x \in \mathbb{R}$ is :
 (a) 7 (b) $(7.2)^{1/7}$ (c) 8 (d) $(3.10)^{1/3}$
79. The value of $\sum_{r=1}^{\infty} \frac{(4r+5)5^{-r}}{r(5r+5)}$ is :
 (a) $\frac{1}{5}$ (b) $\frac{2}{5}$ (c) $\frac{1}{25}$ (d) $\frac{2}{25}$

Answers

1. (c)	2. (c)	3. (b)	4. (b)	5. (a)	6. (b)	7. (d)	8. (a)	9. (a)	10. (c)
11. (d)	12. (c)	13. (b)	14. (c)	15. (b)	16. (c)	17. (a)	18. (c)	19. (c)	20. (d)
21. (b)	22. (d)	23. (d)	24. (a)	25. (a)	26. (b)	27. (d)	28. (b)	29. (d)	30. (d)
31. (a)	32. (d)	33. (c)	34. (c)	35. (b)	36. (d)	37. (b)	38. (d)	39. (d)	40. (a)
41. (b)	42. (b)	43. (b)	44. (a)	45. (c)	46. (a)	47. (c)	48. (d)	49. (c)	50. (b)
51. (b)	52. (a)	53. (b)	54. (b)	55. (b)	56. (a)	57. (d)	58. (b)	59. (d)	60. (c)
61. (d)	62. (b)	63. (c)	64. (c)	65. (d)	66. (d)	67. (d)	68. (a)	69. (d)	70. (c)
71. (a)	72. (d)	73. (d)	74. (a)	75. (b)	76. (b)	77. (c)	78. (c)	79. (a)	

Exercise-2 : One or More than One Answer is/are Correct

- If the first and $(2n - 1)^{\text{th}}$ terms of an A.P, G.P and H.P with positive terms are equal and their n^{th} terms are a , b and c respectively, then which of the following options must be correct :
 - $a + c = 2b$
 - $a \geq b \geq c$
 - $\frac{2ac}{a+c} = b$
 - $ac = b^2$
- Let a, b, c are distinct real numbers such that expression $ax^2 + bx + c$, $bx^2 + cx + a$ and $cx^2 + ax + b$ are always positive then possible value(s) of $\frac{a^2 + b^2 + c^2}{ab + bc + ca}$ may be :
 - 1
 - 2
 - 3
 - 4
- If a, b, c are in H.P, where $a > c > 0$, then :
 - $b > \frac{a+c}{2}$
 - $\frac{1}{a-b} - \frac{1}{b-c} < 0$
 - $ac > b^2$
 - $bc(1-a), ac(1-b), ab(1-c)$ are in A.P
- In an A.P, let T_r denote r^{th} term from beginning, $T_p = \frac{1}{q(p+q)}, T_q = \frac{1}{p(p+q)}$, then :
 - $T_1 = \text{common difference}$
 - $T_{p+q} = \frac{1}{pq}$
 - $T_{pq} = \frac{1}{p+q}$
 - $T_{p+q} = \frac{1}{p^2q^2}$
- Which of the following statement(s) is(are) correct?
 - Sum of the reciprocal of all the n harmonic means inserted between a and b is equal to n times the harmonic mean between two given numbers a and b .
 - Sum of the cubes of first n natural number is equal to square of the sum of the first n natural numbers.
 - If $a, A_1, A_2, A_3, \dots, A_{2n}, b$ are in A.P then $\sum_{i=1}^{2n} A_i = n(a+b)$.
 - If the first term of the geometric progression $g_1, g_2, g_3, \dots, \infty$ is unity, then the value of the common ratio of the progression such that $(4g_2 + 5g_3)$ is minimum equals $\frac{2}{5}$.
- If a, b, c are in 3 distinct numbers in H.P, $a, b, c > 0$, then :
 - $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P
 - $\frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ are in A.P
 - $a^5 + c^5 \geq 2b^5$
 - $\frac{a-b}{b-c} = \frac{a}{c}$

7. All roots of equation $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$ are in G.P. If the sum of their reciprocals is 10, then δ can be equal to :
- (a) 32 (b) -32 (c) $\frac{1}{32}$ (d) $-\frac{1}{32}$
8. Let $a_1, a_2, a_3, \dots$ be a sequence of non-zero real numbers which are in A.P. for $k \in N$. Let $f_k(x) = a_k x^2 + 2a_{k+1}x + a_{k+2}$
- (a) $f_k(x) = 0$ has real roots for each $k \in N$.
 (b) Each of $f_k(x) = 0$ has one root in common.
 (c) Non-common roots of $f_1(x) = 0, f_2(x) = 0, f_3(x) = 0, \dots$ form an A.P.
 (d) None of these
9. Given a, b, c are in A.P., b, c, d are in G.P. and c, d, e are in H.P. If $a = 2$ and $e = 18$, then the possible value of 'c' can be :
- (a) 9 (b) -6 (c) 6 (d) -9
10. The number a, b, c in that order form a three term A.P. and $a + b + c = 60$. The number $(a - 2), b, (c + 3)$ in that order form a three term G.P. All possible values of $(a^2 + b^2 + c^2)$ is/are :
- (a) 1218 (b) 1208 (c) 1288 (d) 1298
11. If $(x^2 + x + 1) + (x^2 + 2x + 3) + (x^2 + 3x + 5) + \dots + (x^2 + 20x + 39) = 4500$, then x is equal to :
- (a) 10 (b) -10 (c) 20.5 (d) -20.5
12. For ΔABC , if $81 + 144a^4 + 16b^4 + 9c^4 = 144abc$, (where notations have their usual meaning), then :
- (a) $a > b > c$ (b) $A < B < C$
 (c) Area of $\Delta ABC = \frac{3\sqrt{3}}{8}$ (d) Triangle ABC is right angled
13. Let $x, y, z \in \left(0, \frac{\pi}{2}\right)$ are first three consecutive terms of an arithmetic progression such that $\cos x + \cos y + \cos z = 1$ and $\sin x + \sin y + \sin z = \frac{1}{\sqrt{2}}$, then which of the following is/are correct ?
- (a) $\cot y = \sqrt{2}$ (b) $\cos(x - y) = \frac{\sqrt{3} - \sqrt{2}}{2\sqrt{2}}$
 (c) $\tan 2y = \frac{2\sqrt{2}}{3}$ (d) $\sin(x - y) + \sin(y - z) = 0$
14. If the numbers 16, 20, 16, d form a A.G.P., then d can be equal to.
- (a) 3 (b) 11 (c) -8 (d) -16

15. Given $\frac{\overbrace{1000\dots 01}^{n \text{ zeroes}}}{\underbrace{1000\dots 01}_{(n+1) \text{ zeroes}}} < \frac{\overbrace{1000\dots 01}^{m \text{ zeroes}}}{\underbrace{1000\dots 01}_{(m+1) \text{ zeroes}}}$, then which of the following is true
- (a) $m + 1 < n$ (b) $m < n$ (c) $m < n + 1$ (d) $m > n + 1$
16. If $S_r = \sqrt{r + \sqrt{r + \sqrt{r + \sqrt{r + \sqrt{\dots \infty}}}}}$, $r > 0$, then which of the following is/are correct.
- (a) S_2, S_6, S_{12}, S_{20} are in A.P. (b) S_4, S_9, S_{16} are irrational
(c) $(2S_3 - 1)^2, (2S_4 - 1)^2, (2S_5 - 1)^2$ are in A.P. (d) S_2, S_{12}, S_{56} are in G.P.
17. Consider the A.P. 50, 48, 46, 44, If S_n denotes the sum to n terms of this A.P., then
- (a) S_n is maximum for $n = 25$ (b) the first negative terms is 26^{th} term
(c) the first negative term is 27^{th} term (d) the maximum value of S_n is 650
18. Let S_n be the sum to n terms of the series $\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \frac{9}{1^2 + 2^2 + 3^2 + 4^2} + \dots$ then
- (a) $S_5 = 5$ (b) $S_{50} = \frac{100}{17}$ (c) $S_{1001} = \frac{1001}{97}$ (d) $S_{\infty} = 6$
19. For ΔABC , if $81 + 144a^4 + 16b^4 + 9c^4 = 144abc$, (where notations have their usual meaning), then
- (a) $a > b > c$ (b) $A < B < C$
(c) Area of $\Delta ABC = \frac{3\sqrt{3}}{8}$ (d) Triangle ABC is right angled

Answers

1.	(b, d)	2.	(b, c)	3.	(b, c, d)	4.	(a, b, c)	5.	(b, c)	6.	(a, b, c, d)
7.	(a, b)	8.	(a, b)	9.	(b, c)	10.	(b, d)	11.	(a, d)	12.	(b, c, d)
13.	(a, b)	14.	(b)	15.	(b, c)	16.	(a, b, c, d)	17.	(a, c, d)	18.	(a, b, d)
19.	(b, c, d)										

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

The first four terms of a sequence are given by $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$.

The general term is given by $T_n = A\alpha^{n-1} + B\beta^{n-1}$ where A, B, α, β are independent of n and A is positive.

1. The value of $(\alpha^2 + \beta^2 + \alpha\beta)$ is equal to :
 (a) 1 (b) 2 (c) 5 (d) 4
2. The value of $5(A^2 + B^2)$ is equal to :
 (a) 2 (b) 4 (c) 6 (d) 8

Paragraph for Question Nos. 3 to 4

There are two sets A and B each of which consists of three numbers in A.P whose sum is 15. D and d are their respective common differences such that $D - d = 1, D > 0$. If $\frac{p}{q} = \frac{7}{8}$ where p and q are the product of the numbers in those sets A and B respectively.

3. Sum of the product of the numbers in set A taken two at a time is :
 (a) 51 (b) 71 (c) 74 (d) 86
4. Sum of the product of the numbers in set B taken two at a time is :
 (a) 52 (b) 54 (c) 64 (d) 74

Paragraph for Question Nos. 5 to 7

Let x, y, z are positive reals and $x + y + z = 60$ and $x > 3$.

5. Maximum value of $(x - 3)(y + 1)(z + 5)$ is :
 (a) (17) (21) (25) (b) (20) (21) (23) (c) (21) (21) (21) (d) (23) (19) (15)
6. Maximum value of $(x - 3)(2y + 1)(3z + 5)$ is :
 (a) $\frac{(355)^3}{3^3 \cdot 6^2}$ (b) $\frac{(355)^3}{3^3 \cdot 6^3}$ (c) $\frac{(355)^3}{3^2 \cdot 6^3}$ (d) None of these
7. Maximum value of xyz is :
 (a) 8×10^3 (b) 27×10^3 (c) 64×10^3 (d) 125×10^3

Paragraph for Question Nos. 8 to 10

Two consecutive numbers from n natural numbers $1, 2, 3, \dots, n$ are removed. Arithmetic mean of the remaining numbers is $\frac{105}{4}$.

8. The value of n is :
 (a) 48 (b) 50 (c) 52 (d) 49
9. The G.M. of the removed numbers is :
 (a) $\sqrt{30}$ (b) $\sqrt{42}$ (c) $\sqrt{56}$ (d) $\sqrt{72}$
10. Let removed numbers are x_1, x_2 then $x_1 + x_2 + n =$
 (a) 61 (b) 63 (c) 65 (d) 69

Paragraph for Question Nos. 11 to 13

The sequence $\{a_n\}$ is defined by formula $a_0 = 4$ and $a_{n+1} = a_n^2 - 2a_n + 2$ for $n \geq 0$. Let the sequence $\{b_n\}$ is defined by formula $b_0 = \frac{1}{2}$ and $b_n = \frac{2a_0 a_1 a_2 \dots a_{n-1}}{a_n} \forall n \geq 1$.

11. The value of a_{10} is equal to :
 (a) $1 + 2^{1024}$ (b) 4^{1024} (c) $1 + 3^{1024}$ (d) 6^{1024}
12. The value of n for which $b_n = \frac{3280}{3281}$ is :
 (a) 2 (b) 3 (c) 4 (d) 5
13. The sequence $\{b_n\}$ satisfies the recurrence formula :
 (a) $b_{n+1} = \frac{2b_n}{1 - b_n^2}$ (b) $b_{n+1} = \frac{2b_n}{1 + b_n^2}$
 (c) $\frac{b_n}{1 + 2b_n^2}$ (d) $\frac{b_n}{1 - 2b_n^2}$

Paragraph for Question Nos. 14 to 15

Let $f(n) = \sum_{r=2}^n \frac{r}{r C_2} \cdot \frac{r+1}{r+1 C_2}$, $a = \lim_{n \rightarrow \infty} f(n)$ and $x^2 - \left(2a - \frac{1}{2}\right)x + t = 0$ has two positive roots α and β .

14. If value of $f(7) + f(8)$ is $\frac{p}{q}$ where p and q are relatively prime, then $(p - q)$ is :
 (a) 53 (b) 55 (c) 57 (d) 59
15. Minimum value of $\frac{4}{\alpha} + \frac{1}{\beta}$ is :
 (a) 2 (b) 6 (c) 3 (d) 4

Paragraph for Question Nos. 16 to 17

Given the sequence of number $a_1, a_2, a_3, \dots, a_{1005}$

which satisfy $\frac{a_1}{a_1 + 1} = \frac{a_2}{a_2 + 3} = \frac{a_3}{a_3 + 5} = \dots = \frac{a_{1005}}{a_{1005} + 2009}$

Also $a_1 + a_2 + a_3 + \dots + a_{1005} = 2010$

16. Nature of the sequence is :

(a) A.P

(b) G.P

(c) A.G.P

(d) H.P

17. 21st term of the sequence is equal to :

(a) $\frac{86}{1005}$

(b) $\frac{83}{1005}$

(c) $\frac{82}{1005}$

(d) $\frac{79}{1005}$

Answers

1. (b)	2. (a)	3. (b)	4. (d)	5. (c)	6. (a)	7. (a)	8. (b)	9. (c)	10. (c)
11. (c)	12. (b)	13. (b)	14. (d)	15. (b)	16. (a)	17. (c)			

Exercise-4 : Matching Type Problems

1.

	Column-I		Column-II
(A)	If three unequal numbers a, b, c are in A.P. and $b - a, c - b, a$ are in G.P, then $\frac{a^3 + b^3 + c^3}{3abc}$ is equal to	(P)	1
(B)	Let x be the arithmetic mean and y, z be two geometric means between any two positive numbers, then $\frac{y^3 + z^3}{2xyz}$ is equal to	(Q)	4
(C)	If a, b, c be three positive number which form three successive terms of a G.P and $c > 4b - 3a$, then the common ratio of the G.P can be equal to	(R)	2
(D)	Number of integral values of x satisfying inequality, $-7x^2 + 8x - 9 > 0$ is	(S)	0

2.

	Column-I		Column-II
(A)	The sequence $a, b, 10, c, d$ are in A.P, then $a + b + c + d =$	(P)	6
(B)	Six G.M.'s are inserted between 2 and 5, if their product can be expressed as $(10)^n$. Then $n =$	(Q)	2
(C)	Let $a_1, a_2, a_3, \dots, a_{10}$ are in A.P and $h_1, h_2, h_3, \dots, h_{10}$ are in H.P such that $a_1 = h_1 = 1$ and $a_{10} = h_{10} = 6$, then $a_4 h_7 =$	(R)	3
(D)	If $\log_3 2, \log_3 (2^x - 5)$ and $\log_3 \left(2^x - \frac{7}{2}\right)$ are in A.P, then $x =$	(S)	20
		(T)	40

3.

	Column-I		Column-II
(A)	The number of real values of x such that three numbers $2^x, 2^{x^2}$ and 2^{x^3} form a non-constant arithmetic progression in that order, is	(P)	0
(B)	Let $S = (a_2 - a_3) \left(\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} \right)$ where $a_1, a_2, a_3, \dots, a_n$ are n consecutive terms of an A.P. and $a_i > 0 \forall i \in \{1, 2, \dots, n\}$. If $a_1 = 225, a_n = 400$, then the value of $S + 7$ is equal to	(Q)	1

(C)	Let S_n denote the sum of first n terms of an non constant A.P and $S_{2n} = 3S_n$, then $\frac{S_{3n}}{2S_n}$ is equal to	(R)	2
(D)	If t_1, t_2, t_3, t_4 and t_5 are first 5 terms of an A.P, then $\frac{4(t_1 - t_2 - t_4) + 6t_3 + t_5}{3t_1}$ is equal to	(S)	3
		(T)	4

4. Column-I contains S and Column-II gives last digit of S.

	Column-I		Column-II
(A)	$S = \sum_{n=1}^{11} (2n-1)^2$	(P)	0
(B)	$S = \sum_{n=1}^{10} (2n-1)^3$	(Q)	1
(C)	$S = \sum_{n=1}^{18} (2n-1)^2 (-1)^n$	(R)	3
(D)	$S = \sum_{n=1}^{15} (2n-1)^3 (-1)^{n-1}$	(S)	5
		(T)	8

5.

	Column-I		Column-II
(A)	If $x, y \in R^+$ satisfy $\log_8 x + \log_4 y^2 = 5$ and $\log_8 y + \log_4 x^2 = 7$ then the value of $\frac{x^2 + y^2}{2080} =$	(P)	6
(B)	In $\triangle ABC$ A, B, C are in A.P and sides a, b and c are in G.P then $a^2(b-c) + b^2(c-a) + c^2(a-b) =$	(Q)	3
(C)	If a, b, c are three positive real numbers then the minimum value of $\frac{b+c}{a} + \frac{a+c}{b} + \frac{a+b}{c}$ is	(R)	0
(D)	In $\triangle ABC$, $(a+b+c)(b+c-a) = \lambda bc$ where $\lambda \in I$, then greatest value of λ is	(S)	2

6. Let $f(n) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n}$ such that $P(n)f(n+2) = P(n)f(n) + q(n)$. Where $P(n)$, $Q(n)$ are polynomials of least possible degree and $P(n)$ has leading coefficient unity. Then match the following Column-I with Column-II.

Column-I		Column-II	
(A)	$\sum_{n=1}^m \frac{p(n)-2}{n}$	(P)	$\frac{m(m+1)}{2}$
(B)	$\sum_{n=1}^m \frac{q(n)-3}{2}$	(Q)	$\frac{5m(m+7)}{2}$
(C)	$\sum_{n=1}^m \frac{p(n)+q^2(n)-11}{n}$	(R)	$\frac{3m(m+7)}{2}$
(D)	$\sum_{n=1}^m \frac{q^2(n)-p(n)-7}{n}$	(S)	$\frac{m(m+7)}{2}$

Answers

1. $A \rightarrow R$; $B \rightarrow P$; $C \rightarrow Q$; $D \rightarrow S$
2. $A \rightarrow R$, $B \rightarrow R$, $C \rightarrow P$, $D \rightarrow R$
3. $A \rightarrow P$, $B \rightarrow R$, $C \rightarrow S$, $D \rightarrow Q$
4. $A \rightarrow Q$; $B \rightarrow P$; $C \rightarrow T$; $D \rightarrow S$
5. $A \rightarrow S$; $B \rightarrow R$; $C \rightarrow P$; $D \rightarrow Q$
6. $A \rightarrow S$; $B \rightarrow P$; $C \rightarrow Q$; $D \rightarrow R$

Exercise-5 : Subjective Type Problems

- Let a, b, c, d are four distinct consecutive numbers in A.P. The complete set of values of x for which $2(a-b) + x(b-c)^2 + (c-a)^3 = 2(a-d) + (b-d)^2 + (c-d)^3$ is true is $(-\infty, \alpha] \cup [\beta, \infty)$, then $|\alpha|$ is equal to :
- The sum of all digits of n for which $\sum_{r=1}^n r 2^r = 2 + 2^{n+10}$ is :
- If $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r+2}{2^{r+1} r(r+1)} = \frac{1}{k}$, then $k =$
- The value of $\sum_{r=1}^{\infty} \frac{8r}{4r^4 + 1}$ is equal to :
- Three distinct non-zero real numbers form an A.P. and the squares of these numbers taken in same order form a G.P. If possible common ratio of G.P. are $3 \pm \sqrt{n}$, $n \in \mathbb{N}$ then $n =$
- If $\sqrt{\underbrace{(1111 \dots 1)}_{2n \text{ times}} - \underbrace{(222 \dots 2)}_{n \text{ times}}} = \underbrace{PPP \dots P}_{n \text{ times}}$ then $P =$
- In an increasing sequence of four positive integers, the first 3 terms are in A.P., the last 3 terms are in G.P. and the fourth term exceed the first term by 30, then the common difference of A.P. lying in interval $[1, 9]$ is :
- The limit of $\frac{1}{n^4} \sum_{k=1}^n k(k+2)(k+4)$ as $n \rightarrow \infty$ is equal to $\frac{1}{\lambda}$, then $\lambda =$
- What is the last digit of $1 + 2 + 3 + \dots + n$ if the last digit of $1^3 + 2^3 + \dots + n^3$ is 1?
- Three distinct positive numbers a, b, c are in G.P., while $\log_c a, \log_b c, \log_a b$ are in A.P. with non-zero common difference d , then $2d =$
- The numbers $\frac{1}{3}, \frac{1}{3} \log_x y, \frac{1}{3} \log_y z, \frac{1}{7} \log_z x$ are in H.P. If $y = x^r$ and $z = x^s$, then $4(r+s) =$
- If $\sum_{k=1}^{\infty} \frac{k^2}{3^k} = \frac{p}{q}$; where p and q are relatively prime positive integers. Find the value of $(p+q)$.
- The sum of the terms of an infinitely decreasing Geometric Progression (GP) is equal to the greatest value of the function $f(x) = x^3 + 3x - 9$ when $x \in [-4, 3]$ and the difference between the first and second term is $f'(0)$. The common ratio $r = \frac{p}{q}$ where p and q are relatively prime positive integers. Find $(p+q)$.
- A cricketer has to score 4500 runs. Let a_n denotes the number of runs he scores in the n^{th} match. If $a_1 = a_2 = \dots a_{10} = 150$ and $a_{10}, a_{11}, a_{12}, \dots$ are in A.P. with common difference (-2) . If N be the total number of matches played by him to score 4500 runs. Find the sum of the digits of N .

15. If $x = 10 \sum_{n=3}^{100} \frac{1}{n^2 - 4}$, then $[x] =$ (where $[.]$ denotes greatest integer function)
16. Let $f(n) = \frac{4n + \sqrt{4n^2 - 1}}{\sqrt{2n+1} + \sqrt{2n-1}}$, $n \in N$ then the remainder when $f(1) + f(2) + f(3) + \dots + f(60)$ is divided by 9 is.
17. Find the sum of series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \frac{1}{9} + \frac{1}{12} + \dots \infty$, where the terms are the reciprocals of the positive integers whose only prime factors are two's and three's :
18. Let $a_1, a_2, a_3, \dots, a_n$ be real numbers in arithmetic progression such that $a_1 = 15$ and a_2 is an integer. Given $\sum_{r=1}^{10} (a_r)^2 = 1185$. If $S_n = \sum_{r=1}^n a_r$ and maximum value of n is N for which $S_n \geq S_{(n-1)}$, then find $N - 10$.
19. Let the roots of the equation $24x^3 - 14x^2 + kx + 3 = 0$ form a geometric sequence of real numbers. If absolute value of k lies between the roots of the equation $x^2 + \alpha^2 x - 112 = 0$, then the largest integral value of α is :
20. How many ordered pair(s) satisfy $\log \left(x^3 + \frac{1}{3}y^3 + \frac{1}{9} \right) = \log x + \log y$
21. Let a and b be positive integers. The value of xyz is 55 and $\frac{343}{55}$ when a, x, y, z, b are in arithmetic and harmonic progression respectively. Find the value of $(a + b)$

Answers

1.	8	2.	9	3.	2	4.	2	5.	8	6.	3	7.	9
8.	4	9.	1	10.	3	11.	6	12.	5	13.	5	14.	7
15.	5	16.	8	17.	3	18.	6	19.	2	20.	1	21.	8



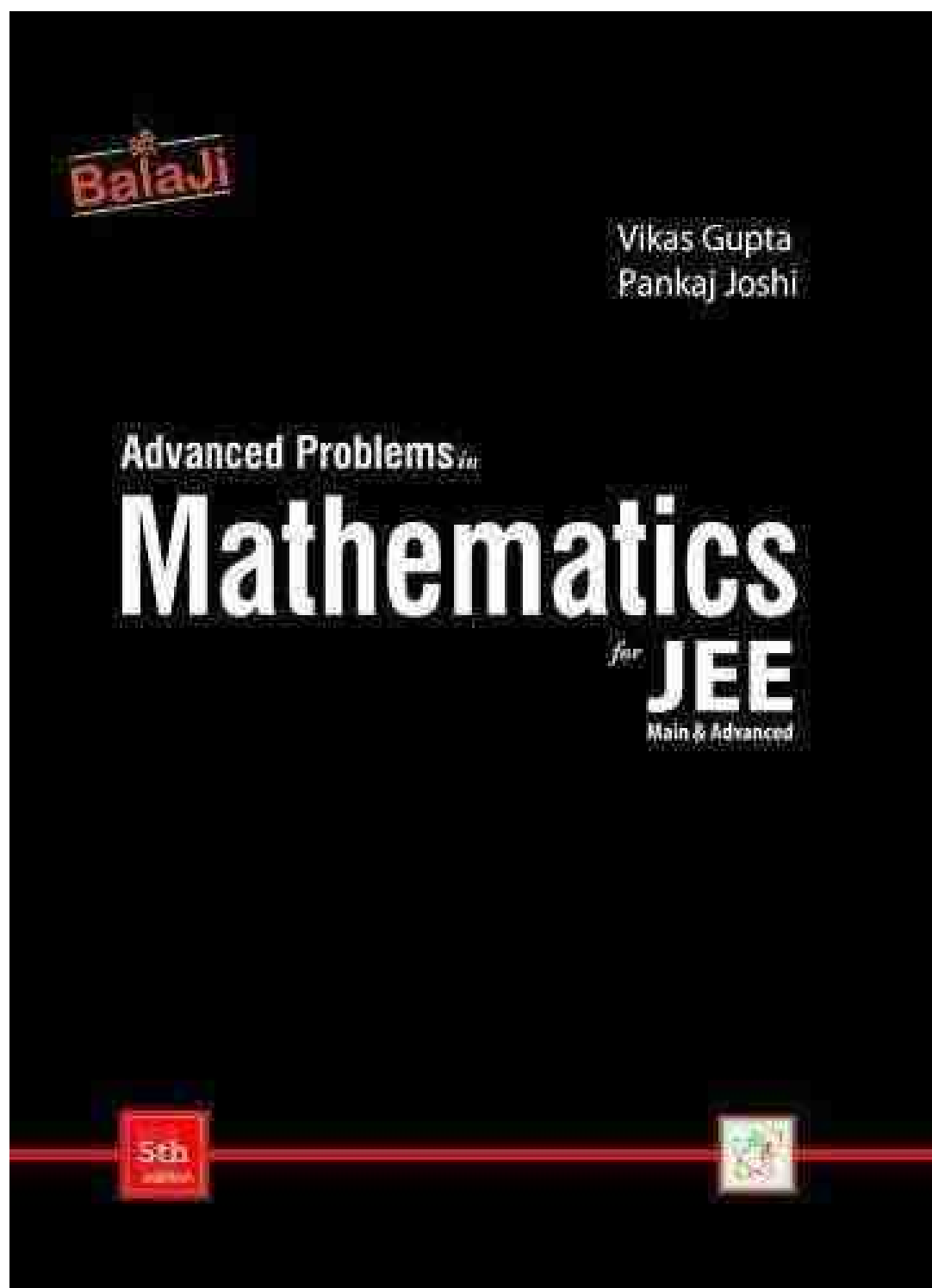
Balaji

Advanced Problems in Mathematics Chapter 10 to 26

for IIT JEE Main and Advanced

by

Vikas Gupta and Pankaj Joshi



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BalaJi

Advanced Problems *in*
MATHEMATICS

for
JEE (MAIN & ADVANCED)

by :

Vikas Gupta
Director
Vibrant Academy India(P) Ltd.
KOTA (Rajasthan)

Pankaj Joshi
Director
Vibrant Academy India(P) Ltd.
KOTA (Rajasthan)

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Exercise-1 : Single Choice Problems

1. If $\begin{vmatrix} 1 & \cos \alpha & \cos \beta \\ \cos \alpha & 1 & \cos \gamma \\ \cos \beta & \cos \gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos \alpha & \cos \beta \\ \cos \alpha & 0 & \cos \gamma \\ \cos \beta & \cos \gamma & 0 \end{vmatrix}$ then the value of $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$ is :
 (a) 1 (b) $\frac{3}{2}$ (c) $\frac{3}{8}$ (d) $\frac{9}{4}$
2. Let the following system of equations

$$\begin{aligned} kx + y + z &= 1 \\ x + ky + z &= k \\ x + y + kz &= k^2 \end{aligned}$$
 has no solution. Find $|k|$.
 (a) 0 (b) 1 (c) 2 (d) 3
3. If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and vectors $(1, a, a^2)$, $(1, b, b^2)$ and $(1, c, c^2)$ are non-coplanar, then the product abc equals :
 (a) 2 (b) -1 (c) 1 (d) 0
4. If the system of linear equations

$$\begin{aligned} x + 2ay + az &= 0 \\ x + 3by + bz &= 0 \\ x + 4cy + cz &= 0 \end{aligned}$$
 has a non-zero solution, then a, b, c :
 (a) are in A.P. (b) are in G.P.
 (c) are in H.P. (d) satisfy $a + 2b + 3c = 0$
5. If the number of quadratic polynomials $ax^2 + 2bx + c$ which satisfy the following conditions :
 (i) a, b, c are distinct

- (ii) $a, b, c \in \{1, 2, 3, \dots, 2001, 2002\}$
 (iii) $x + 1$ divides $ax^2 + 2bx + c$
 is equal to 1000λ , then find the value of λ .
 (a) 2002 (b) 2001 (c) 2003 (d) 2004
6. If the system of equations $2x + ay + 6z = 8$, $x + 2y + z = 5$, $2x + ay + 3z = 4$ has a unique solution then 'a' cannot be equal to :
 (a) 2 (b) 3 (c) 4 (d) 5
7. If one of the roots of the equation $\begin{vmatrix} 7 & 6 & x^2 - 13 \\ 2 & x^2 - 13 & 2 \\ x^2 - 13 & 3 & 7 \end{vmatrix} = 0$ is $x = 2$, then sum of all other five roots is :
 (a) -2 (b) 0 (c) $2\sqrt{5}$ (d) $\sqrt{15}$
8. The system of equations

$$\begin{aligned} kx + (k+1)y + (k-1)z &= 0 \\ (k+1)x + ky + (k+2)z &= 0 \\ (k-1)x + (k+2)y + kz &= 0 \end{aligned}$$

 has a nontrivial solution for :
 (a) Exactly three real values of k . (b) Exactly two real values of k .
 (c) Exactly one real value of k . (d) Infinite number of values of k .
9. If $a_1, a_2, a_3, \dots, a_n$ are in G.P. and $a_i > 0$ for each i , then the determinant $\Delta = \begin{vmatrix} \log a_n & \log a_{n+2} & \log a_{n+4} \\ \log a_{n+6} & \log a_{n+8} & \log a_{n+10} \\ \log a_{n+12} & \log a_{n+14} & \log a_{n+16} \end{vmatrix}$ is equal to :
 (a) 0 (b) $\log \left(\sum_{i=1}^{n^2+n} a_i \right)$ (c) 1 (d) 2
10. If $D_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D_2 = \begin{vmatrix} a_1 + 2a_2 + 3a_3 & 2a_3 & 5a_2 \\ b_1 + 2b_2 + 3b_3 & 2b_3 & 5b_2 \\ c_1 + 2c_2 + 3c_3 & 2c_3 & 5c_2 \end{vmatrix}$ then $\frac{D_2}{D_1}$ is equal to :
 (a) 10 (b) -10 (c) 20 (d) -20
11. If $\Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} 1 & bc & a \\ 1 & ac & b \\ 1 & ab & c \end{vmatrix}$ then :
 (a) $\Delta_1 = \Delta_2$ (b) $\Delta_1 = 2\Delta_2$ (c) $\Delta_1 + \Delta_2 = 0$ (d) $\Delta_1 + 2\Delta_2 = 0$
12. The value of the determinant $\begin{vmatrix} 1 & 0 & -1 \\ a & 1 & 1-a \\ b & a & 1+a-b \end{vmatrix}$ depends on :
 (a) only a (b) only b (c) neither a nor b (d) both a and b

13. Sum of solutions of the equation $\begin{vmatrix} 1 & 2 & x \\ 2 & 3 & x^2 \\ 3 & 5 & 2 \end{vmatrix} = 10$ is :
- (a) 1 (b) -1 (c) 2 (d) 4
14. If $D = \begin{vmatrix} x+d & x+e & x+f \\ x+d+1 & x+e+1 & x+f+1 \\ x+a & x+b & x+c \end{vmatrix}$ then D does not depend on :
- (a) a (b) e (c) d (d) x
15. The value of the determinant $\begin{vmatrix} x-y-z & 2x & 2x \\ 2y & y-z-x & 2y \\ 2z & 2z & z-x-y \end{vmatrix} =$
- (a) $xyz(x+y+z)^2$ (b) $(x+y-z)(x+y+z)^2$
 (c) $(x+y+z)^3$ (d) $(x+y+z)^2$
16. A rectangle $ABCD$ is inscribed in a circle. Let PQ be the diameter of the circle parallel to the side AB . If $\angle BPC = 30^\circ$, then the ratio of the area of rectangle to the area of circle is :
- (a) $\frac{\sqrt{3}}{\pi}$ (b) $\frac{\sqrt{3}}{2\pi}$ (c) $\frac{3}{\pi}$ (d) $\frac{\sqrt{3}}{9\pi}$
17. Let $ab = 1$, $\Delta = \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$ then the minimum value of Δ is :
- (a) 3 (b) 9 (c) 27 (d) 81
18. The determinant $\begin{vmatrix} 2 & a+b+c+d & ab+cd \\ a+b+c+d & 2(a+b)(c+d) & ab(c+d)+cd(a+b) \\ ab+cd & ab(c+d)+cd(a+b) & 2abcd \end{vmatrix} = 0$ for
- (a) $a+b+c+d=0$ (b) $ab+cd=0$
 (c) $ab(c+d)+cd(a+b)=0$ (d) any a, b, c, d
19. Let $\det A = \begin{vmatrix} l & m & n \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$ and if $(l-m)^2 + (p-q)^2 = 9$, $(m-n)^2 + (q-r)^2 = 16$, $(n-l)^2 + (r-p)^2 = 25$, then the value of $(\det A)^2$ equals :
- (a) 36 (b) 100 (c) 144 (d) 169
20. The number of distinct real values of K such that the system of equations $x+2y+z=1$, $x+3y+4z=K$, $x+5y+10z=K^2$ has infinitely many solutions is :
- (a) 0 (b) 4 (c) 2 (d) 3

21. If $\begin{vmatrix} (x+1) & (x+1)^2 & (x+1)^3 \\ (x+2) & (x+2)^2 & (x+2)^3 \\ (x+3) & (x+3)^2 & (x+3)^3 \end{vmatrix}$ is expressed as a polynomial in x , then the term independent of x is :
- (a) 0 (b) 2 (c) 12 (d) 16
22. If A, B, C are the angles of triangle ABC , then the minimum value of $\begin{vmatrix} -2 & \cos C & \cos B \\ \cos C & -1 & \cos A \\ \cos B & \cos A & -1 \end{vmatrix}$ is equal to :
- (a) 0 (b) -1 (c) 1 (d) -2
23. If the system of linear equations
- $$\begin{aligned} x + 2ay + az &= 0 \\ x + 3by + bz &= 0 \\ x + 4cy + cz &= 0 \end{aligned}$$
- has a non-zero solution then a, b, c are in
- (a) A.P. (b) G.P. (c) H.P. (d) None of these
24. If a, b and c are the roots of the equation $x^3 + 2x^2 + 1 = 0$, find $\begin{vmatrix} a & b & x \\ b & c & a \\ c & a & b \end{vmatrix}$.
- (a) 8 (b) -8 (c) 0 (d) 2
25. The system of homogeneous equation $\lambda x + (\lambda + 1)y + (\lambda - 1)z = 0$, $(\lambda + 1)x + \lambda y + (\lambda + 2)z = 0$, $(\lambda - 1)x + (\lambda + 2)y + \lambda z = 0$ has non-trivial solution for :
- (a) exactly three real values of λ (b) exactly two real values of λ
 (c) exactly three real value of λ (d) infinitely many real value of λ
26. If one of the roots of the equation $\begin{vmatrix} 7 & 6 & x^2 - 13 \\ 2 & x^2 - 13 & 2 \\ x^2 - 13 & 3 & 7 \end{vmatrix} = 0$ is $x = 2$, then sum of all other five roots is :
- (a) -2 (b) 0 (c) $2\sqrt{5}$ (d) $\sqrt{15}$

Answers

1. (a)	2. (c)	3. (b)	4. (c)	5. (a)	6. (c)	7. (a)	8. (c)	9. (a)	10. (b)
11. (c)	12. (c)	13. (b)	14. (d)	15. (c)	16. (a)	17. (c)	18. (d)	19. (c)	20. (c)
21. (c)	22. (b)	23. (c)	24. (a)	25. (c)	26. (a)				

Exercise-2 : One or More than One Answer is/are Correct

1. Let $f(a, b) = \begin{vmatrix} a & a^2 & 0 \\ 1 & (2a+b) & (a+b)^2 \\ 0 & 1 & (2a+3b) \end{vmatrix}$, then

- (a) $(2a+b)$ is a factor of $f(a, b)$ (b) $(a+2b)$ is a factor of $f(a, b)$
 (c) $(a+b)$ is a factor of $f(a, b)$ (d) a is a factor of $f(a, b)$

2. If $\begin{vmatrix} 1+\cos^2\theta & \sin^2\theta & 2\sqrt{3}\tan\theta \\ \cos^2\theta & 1+\sin^2\theta & 2\sqrt{3}\tan\theta \\ \cos^2\theta & \sin^2\theta & 1+2\sqrt{3}\tan\theta \end{vmatrix} = 0$ then θ may be :

- (a) $\frac{\pi}{6}$ (b) $\frac{5\pi}{6}$ (c) $\frac{7\pi}{6}$ (d) $\frac{11\pi}{6}$

3. Let $\Delta = \begin{vmatrix} a & a+d & a+3d \\ a+d & a+2d & a \\ a+2d & a & a+d \end{vmatrix}$ then :

- (a) Δ depends on a (b) Δ depends on d
 (c) Δ is independent of a, d (d) $\Delta = 0$

4. The value(s) of λ for which the system of equations

$$(1-\lambda)x + 3y - 4z = 0$$

$$x - (3+\lambda)y + 5z = 0$$

$$3x + y - \lambda z = 0$$

possesses non-trivial solutions.

- (a) -1 (b) 0 (c) 1 (d) 2

5. Let $D(x) = \begin{vmatrix} x^2+4x-3 & 2x+4 & 13 \\ 2x^2+5x-9 & 4x+5 & 26 \\ 8x^2-6x+1 & 16x-6 & 104 \end{vmatrix} = \alpha x^3 + \beta x^2 + \gamma x + \delta$ then :

- (a) $\alpha + \beta = 0$ (b) $\beta + \gamma = 0$ (c) $\alpha + \beta + \gamma + \delta = 0$ (d) $\alpha + \beta + \gamma = 0$

6. Let $D(x) = \begin{vmatrix} x^2+4x-3 & 2x+4 & 13 \\ 2x^2+5x-9 & 4x+5 & 26 \\ 8x^2-6x+1 & 16x-6 & 104 \end{vmatrix} = \alpha x^3 + \beta x^2 + \gamma x + \delta$ then :

- (a) $\alpha + \beta = 0$ (b) $\beta + \gamma = 0$ (c) $\alpha + \beta + \gamma + \delta = 0$ (d) $\alpha + \beta + \gamma = 0$

7. If the system of equations

$$ax + y + 2z = 0$$

$$x + 2y + z = b$$

$$2x + y + az = 0$$

has no solution then $(a+b)$ can be equals to :

- (a) -1 (b) 2 (c) 3 (d) 4

8. If the system of equations

$$ax + y + 2z = 0$$

$$x + 2y + z = b$$

$$2x + y + az = 0$$

has no solution then $(a + b)$ can be equal to

(a) -1

(b) 2

(c) 3

(d) 4

Answers

1.	(b, c, d)	2.	(b, d)	3.	(a, b)	4.	(a, b)	5.	(a, b, d)	6.	(a, b, d)
7.	(b, c, d)	8.	(b)								

Exercise-4 : Subjective Type Problems

1. If 3^n is a factor of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ {}^nC_1 & {}^{n+3}C_1 & {}^{n+6}C_1 \\ {}^nC_2 & {}^{n+3}C_2 & {}^{n+6}C_2 \end{vmatrix}$ then the maximum value of n is

.....

2. Find the value of λ for which $\begin{vmatrix} 2a_1 + b_1 & 2a_2 + b_2 & 2a_3 + b_3 \\ 2b_1 + c_1 & 2b_2 + c_2 & 2b_3 + c_3 \\ 2c_1 + a_1 & 2c_2 + a_2 & 2c_3 + a_3 \end{vmatrix} = \lambda \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

3. Find the co-efficient of x in the expansion of the determinant $\begin{vmatrix} (1+x)^2 & (1+x)^4 & (1+x)^6 \\ (1+x)^3 & (1+x)^6 & (1+x)^9 \\ (1+x)^4 & (1+x)^8 & (1+x)^{12} \end{vmatrix}$.

4. If $x, y, z \in R$ and $\begin{vmatrix} x & y^2 & z^3 \\ x^4 & y^5 & z^6 \\ x^7 & y^8 & z^9 \end{vmatrix} = 2$ then find the value of

$$\begin{vmatrix} y^5 z^6 (z^3 - y^3) & x^4 z^6 (x^3 - z^3) & x^4 y^5 (y^3 - x^3) \\ y^2 z^3 (y^6 - z^6) & xz^3 (z^6 - x^6) & xy^2 (x^6 - y^6) \\ y^2 z^3 (z^3 - y^3) & xz^3 (x^3 - z^3) & xy^2 (y^3 - x^3) \end{vmatrix}.$$

5. If the system of equations :

$$2x + 3y - z = 0$$

$$3x + 2y + kz = 0$$

$$4x + y + z = 0$$

have a set of non-zero integral solutions then, find the smallest positive value of z .

6. Find $a \in R$ for which the system of equations $2ax - 2y + 3z = 0$; $x + ay + 2z = 0$ and $2x + az = 0$ also have a non-trivial solution.

7. If three non-zero distinct real numbers form an arithmetic progression and the squares of these numbers taken in the same order constitute a geometric progression. Find the sum of all possible common ratios of the geometric progression.

8. Let $\Delta_1 = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} 6a_1 & 2a_2 & 2a_3 \\ 3b_1 & b_2 & b_3 \\ 12c_1 & 4c_2 & 4c_3 \end{vmatrix}$ and $\Delta_3 = \begin{vmatrix} 3a_1 + b_1 & 3a_2 + b_2 & 3a_3 + b_3 \\ 3b_1 & 3b_2 & 3b_3 \\ 3c_1 & 3c_2 & 3c_3 \end{vmatrix}$

then $\Delta_3 - \Delta_2 = k\Delta_1$, find k .

9. The minimum value of determinant $\Delta = \begin{vmatrix} 1 & \cos \theta & 1 \\ -\cos \theta & 1 & \cos \theta \\ -1 & -\cos \theta & 2 \end{vmatrix} \forall \theta \in R$ is :

10. For a unique value of μ & λ , the system of equations given by

$$x + y + z = 6$$

$$x + 2y + 3z = 14$$

$$2x + 5y + \lambda z = \mu$$

has infinitely many solutions, then $\frac{\mu - \lambda}{4}$ is equal to

11. Let $\lim_{n \rightarrow \infty} n \sin(2\pi e \lfloor \frac{n}{e} \rfloor) = k\pi$, where $n \in \mathbb{N}$. Find k :

12. If the system of linear equations

$$(\cos \theta)x + (\sin \theta)y + \cos \theta = 0$$

$$(\sin \theta)x + (\cos \theta)y + \sin \theta = 0$$

$$(\cos \theta)x + (\sin \theta)y - \cos \theta = 0$$

is consistent, then the number of possible values of θ , $\theta \in [0, 2\pi]$ is :

Answers

1.	3	2.	9	3.	0	4.	4	5.	5	6.	2	7.	6
8.	3	9.	3	10.	7	11.	2	12.	2				



Exercise-1 : Single Choice Problems

- Let t_1, t_2, t_3 be three distinct points on circle $|t|=1$. If θ_1, θ_2 and θ_3 be the arguments of t_1, t_2, t_3 respectively then $\cos(\theta_1 - \theta_2) + \cos(\theta_2 - \theta_3) + \cos(\theta_3 - \theta_1)$
 - $\geq -\frac{3}{2}$
 - $\leq -\frac{3}{2}$
 - $\geq \frac{3}{2}$
 - ≤ 2
- The number of points of intersection of the curves represented by $\arg(z - 2 - 7i) = \cot^{-1}(2)$ and $\arg\left(\frac{z - 5i}{z + 2 - i}\right) = \pm \frac{\pi}{2}$
 - 0
 - 1
 - 2
 - None of these
- All three roots of $az^3 + bz^2 + cz + d = 0$, have negative real part, ($a, b, c \in \mathbb{R}$) then :
 - All a, b, c, d have the same sign
 - a, b, c have same sign
 - a, b, d have same sign
 - b, c, d have same sign
- Let z_1 and z_2 be two roots of the equation $z^2 + az + b = 0$, z being complex number. Further, assume that the origin, z_1 and z_2 form an equilateral triangle, then :
 - $a^2 = b$
 - $a^2 = 2b$
 - $a^2 = 3b$
 - $a^2 = 4b$
- If z and ω are two non-zero complex numbers such that $|z\omega| = 1$, and $\arg(z) - \arg(\omega) = \frac{\pi}{2}$, then $\bar{z}\omega$ is equal to :
 - 1
 - 1
 - i
 - $-i$
- If ω be an imaginary n^{th} root of unity, then $\sum_{r=1}^n (ar + b)\omega^{r-1}$ is equal to :
 - $\frac{n(n+1)a}{2\omega}$
 - $\frac{nb}{1-n}$
 - $\frac{na}{\omega-1}$
 - None of these

7. If α, β are complex numbers then the maximum value of $\frac{\alpha \bar{\beta} + \bar{\alpha} \beta}{|\alpha \beta|}$ is equal to :
 (a) 1 (b) 2 (c) greater than 2 (d) less than 1
8. Let z_1, z_2, z_3 and z_4 be the roots of the equation $z^4 + z^3 + 2 = 0$, then the value of $\prod_{r=1}^4 (2z_r + 1)$ is equal to :
 (a) 28 (b) 29 (c) 30 (d) 31
9. If $\arg\left(\frac{z-6-3i}{z-3-6i}\right) = \frac{\pi}{4}$, then :
 (a) minimum value of $|z|$ is $6\sqrt{2} - 3$ (b) Maximum value of $|z|$ is $6\sqrt{2} + 3$
 (c) minimum value of $|z|$ is $15\sqrt{2} - 6$ (d) Maximum value of $|z|$ is $15\sqrt{2} + 6$
10. If $z_1 \neq -z_2$ and $|z_1 + z_2| = \left|\frac{1}{z_1} + \frac{1}{z_2}\right|$ then :
 (a) at least one of z_1, z_2 is unimodular (b) both z_1, z_2 are unimodular
 (c) $z_1 \cdot z_2$ is unimodular (d) $z_1 - z_2$ is unimodular
11. If $|z - i| \leq 2$ and $z_1 = 5 + 3i$, then the maximum value of $|iz + z_1|$ is :
 (a) $5 + \sqrt{13}$ (b) $5 + \sqrt{2}$ (c) 7 (d) 8
12. If z_1, z_2, z_3 are vertices of a triangle such that $|z_1 - z_2| = |z_1 - z_3|$ then $\arg\left(\frac{2z_1 - z_2 - z_3}{z_3 - z_2}\right)$ is :
 (a) $\pm \frac{\pi}{3}$ (b) 0 (c) $\pm \frac{\pi}{2}$ (d) $\pm \frac{\pi}{6}$
13. It is given that complex numbers z_1 and z_2 satisfy $|z_1| = 2$ and $|z_2| = 3$. If the included angle of their corresponding vectors is 60° , then $\left|\frac{z_1 + z_2}{z_1 - z_2}\right|$ can be expressed as $\frac{\sqrt{n}}{7}$, where 'n' is a natural number then $n =$
 (a) 126 (b) 119 (c) 133 (d) 19
14. If all the roots of $z^3 + az^2 + bz + c = 0$ are of unit modulus, then :
 (a) $|a| \leq 3$ (b) $|b| \leq 3$ (c) $|c| = 1$ (d) All of the above
15. Let z be a complex number satisfying $\frac{1}{2} \leq |z| \leq 4$, then sum of greatest and least values of $\left|z + \frac{1}{z}\right|$ is :
 (a) $\frac{65}{4}$ (b) $\frac{65}{16}$ (c) $\frac{17}{4}$ (d) 17
16. If $|z - 2i| \leq \sqrt{2}$, then the maximum value of $|3 + i(z - 1)|$ is :
 (a) $\sqrt{2}$ (b) $2\sqrt{2}$ (c) $2 + \sqrt{2}$ (d) $3 + 2\sqrt{2}$

17. Let $x - \frac{1}{x} = (\sqrt{2})i$ where $i = \sqrt{-1}$. Then the value of $x^{2187} - \frac{1}{x^{2187}}$ is :
- (a) $i\sqrt{2}$ (b) $-i\sqrt{2}$ (c) -2 (d) $\frac{i}{\sqrt{2}}$
18. If $z = re^{i\theta}$ ($r > 0$ & $0 \leq \theta < 2\pi$) is a root of the equation $z^8 - z^7 + z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0$ then number of values of ' θ ' is :
- (a) 6 (b) 7 (c) 8 (d) 9
19. Let P and Q be two points on the circle $|w| = r$ represented by w_1 and w_2 respectively, then the complex number representing the point of intersection of the tangents at P and Q is :
- (a) $\frac{w_1 w_2}{2(w_1 + w_2)}$ (b) $\frac{2w_1 \bar{w}_2}{w_1 + w_2}$ (c) $\frac{2w_1 w_2}{w_1 + w_2}$ (d) $\frac{2\bar{w}_1 w_2}{w_1 + w_2}$
20. If z_1, z_2, z_3 are complex number, such that $|z_1| = 2, |z_2| = 3, |z_3| = 4$, then maximum value of $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$ is :
- (a) 58 (b) 29 (c) 87 (d) None of these
21. If $Z = \frac{7+i}{3+4i}$, then find Z^{14} :
- (a) 2^7 (b) $(-2)^7$ (c) $(2^7)i$ (d) $(-2^7)i$
22. If $|Z - 4| + |Z + 4| = 10$, then the difference between the maximum and the minimum values of $|Z|$ is :
- (a) 2 (b) 3 (c) $\sqrt{41} - 5$ (d) 0

Answers

1. (a)	2. (a)	3. (c)	4. (c)	5. (d)	6. (c)	7. (b)	8. (d)	9. (b)	10. (c)
11. (c)	12. (c)	13. (c)	14. (d)	15. (c)	16. (b)	17. (a)	18. (c)	19. (c)	20. (c)
21. (c)	22. (a)								


Exercise-2 : One or More than One Answer is/are Correct

- Let Z_1 and Z_2 are two non-zero complex number such that $|Z_1 + Z_2| = |Z_1| = |Z_2|$, then $\frac{Z_1}{Z_2}$ may be :
 - $1 + \omega$
 - $1 + \omega^2$
 - ω
 - ω^2
- Let z_1, z_2 and z_3 be three distinct complex numbers, satisfying $|z_1| = |z_2| = |z_3| = 1$. Which of the following is/are true :
 - If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then $\arg\left(\frac{z - z_1}{z - z_2}\right) > \frac{\pi}{4}$ where $|z| > 1$
 - $|z_1 z_2 + z_2 z_3 + z_3 z_1| = |z_1 + z_2 + z_3|$
 - $\operatorname{Im}\left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 \cdot z_2 \cdot z_3}\right) = 0$
 - If $|z_1 - z_2| = \sqrt{2}|z_1 - z_3| = \sqrt{2}|z_2 - z_3|$, then $\operatorname{Re}\left(\frac{z_3 - z_1}{z_3 - z_2}\right) = 0$
- The triangle formed by the complex numbers $z, iz, i^2 z$ is :
 - equilateral
 - isosceles
 - right angled
 - isosceles but not right angled
- If $A(z_1), B(z_2), C(z_3), D(z_4)$ lies on $|z| = 4$ (taken in order), where $z_1 + z_2 + z_3 + z_4 = 0$ then:
 - Max. area of quadrilateral $ABCD = 32$
 - Max. area of quadrilateral $ABCD = 16$
 - The triangle $\triangle ABC$ is right angled
 - The quadrilateral $ABCD$ is rectangle
- Let z_1, z_2 and z_3 be three distinct complex numbers satisfying $|z_1| = |z_2| = |z_3| = 1$. Which of the following is/are true ?
 - If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then $\arg\left(\frac{z - z_1}{z - z_2}\right) > \frac{\pi}{4}$ where $|z| > 1$
 - $|z_1 z_2 + z_2 z_3 + z_3 z_1| = |z_1 + z_2 + z_3|$
 - $\operatorname{Im}\left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 \cdot z_2 \cdot z_3}\right) = 0$
 - If $|z_1 - z_2| = \sqrt{2}|z_1 - z_3| = \sqrt{2}|z_2 - z_3|$, then $\operatorname{Re}\left(\frac{z_3 - z_1}{z_3 - z_2}\right) = 0$

6. If $z_1 = a + ib$ and $z_2 = c + id$ are two complex numbers where $a, b, c, d \in \mathbb{R}$ and $|z_1| = |z_2| = 1$ and $\operatorname{Im}(z_1 \bar{z}_2) = 0$. If $w_1 = a + ic$ and $w_2 = b + id$, then :
- (a) $\operatorname{Im}(w_1 \bar{w}_2) = 0$ (b) $\operatorname{Im}(\bar{w}_1 w_2) = 0$
 (c) $\operatorname{Im}\left(\frac{w_1}{w_2}\right) = 0$ (d) $\operatorname{Re}\left(\frac{w_1}{w_2}\right) = 0$
7. The solutions of the equation $z^4 + 4iz^3 - 6z^2 - 4iz - i = 0$ represent vertices of a convex polygon in the complex plane. The area of the polygon is :
 (a) $2^{1/2}$ (b) $2^{3/2}$ (c) $2^{5/2}$ (d) $2^{5/4}$
8. Least positive argument of the 4th root of the complex number $2 - i\sqrt{12}$ is :
 (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{12}$ (c) $\frac{5\pi}{12}$ (d) $\frac{7\pi}{12}$
9. Let ω be the imaginary cube root of unity and $(a + b\omega + c\omega^2)^{2015} = (a + b\omega^2 + c\omega)$ where a, b, c are unequal real numbers. Then the value of $a^2 + b^2 + c^2 - ab - bc - ca$ equals :
 (a) 0 (b) 1 (c) 2 (d) 3
10. Let n be a positive integer and a complex number with unit modulus is a solution of the equation $z^n + z + 1 = 0$ then the value of n can be :
 (a) 62 (b) 155 (c) 221 (d) 196

Answers

1.	(c, d)	2.	(b, c, d)	3.	(b, c)	4.	(a, c, d)	5.	(b, c, d)	6.	(a, b, c)
7.	(d)	8.	(c)	9.	(b)	10.	(a, b, c)				

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

Let $f(z)$ is of the form $\alpha z + \beta$, where α, β are constants and α, β, z are complex numbers such that $|\alpha| \neq |\beta|$. $f(z)$ satisfies following properties :

- (i) If imaginary part of z is non zero, then $f(z) + \overline{f(z)} = f(\bar{z}) + \overline{f(\bar{z})}$
- (ii) If real part of z is zero, then $f(z) + \overline{f(z)} = 0$
- (iii) If z is real, then $\overline{f(z)} f(z) > (z+1)^2 \forall z \in \mathbb{R}$.

1. $\frac{4x^2}{(f(1) - f(-1))^2} + \frac{y^2}{(f(0))^2} = 1, x, y \in \mathbb{R}$, in (x, y) plane will represent :
 (a) hyperbola (b) circle (c) ellipse (d) pair of line
2. Consider ellipse $S : \frac{x^2}{(\operatorname{Re}(\alpha))^2} + \frac{y^2}{(\operatorname{Im}(\beta))^2} = 1, x, y \in \mathbb{R}$ in (x, y) plane, then point $(1, 1)$ will lie :
 (a) outside the ellipse S (b) inside the ellipse S
 (c) on the ellipse S (d) none of these

Paragraph for Question Nos. 3 to 5

Let z_1 and z_2 be complex numbers, such that $z_1^2 - 4z_2 = 16 + 20i$. Also suppose that roots α and β of $t^2 + z_1 t + z_2 + m = 0$ for some complex number m satisfy $|\alpha - \beta| = 2\sqrt{7}$, then :

3. The complex number ' m ' lies on :
 (a) a square with side 7 and centre $(4, 5)$ (b) a circle with radius 7 and centre $(4, 5)$
 (c) a circle with radius 7 and centre $(-4, 5)$ (d) a square with side 7 and centre $(-4, 5)$
4. The greatest value of $|m|$ is :
 (a) $5\sqrt{21}$ (b) $5 + \sqrt{23}$ (c) $7 + \sqrt{43}$ (d) $7 + \sqrt{41}$
5. The least value of $|m|$ is :
 (a) $7 - \sqrt{41}$ (b) $7 - \sqrt{43}$ (c) $5 - \sqrt{23}$ (d) $5 + \sqrt{21}$

Paragraph for Question Nos. 6 to 7

Let $z_1 = 3$ and $z_2 = 7$ represent two points A and B respectively on complex plane. Let the curve C_1 be the locus of point $P(z)$ satisfying $|z - z_1|^2 + |z - z_2|^2 = 10$ and the curve C_2 be the locus of point $P(z)$ satisfying $|z - z_1|^2 + |z - z_2|^2 = 16$.

6. Least distance between curves C_1 and C_2 is :
 (a) 4 (b) 3 (c) 2 (d) 1

7. The locus of point from which tangents drawn to C_1 and C_2 are perpendicular, is :

- (a) $|z - 5| = 4$ (b) $|z - 3| = 2$ (c) $|z - 5| = 3$ (d) $|z - 5| = \sqrt{5}$

Paragraph for Question Nos. 8 to 9

In the Argand plane Z_1, Z_2 and Z_3 are respectively the vertices of an isosceles triangle ABC with $AC = BC$ and $\angle CAB = \theta$. If $I(Z_4)$ is the incentre of triangle, then :

8. The value of $\left(\frac{AB}{IA}\right)^2 \left(\frac{AC}{AB}\right)$ is equal to :

- (a) $\left| \frac{(Z_2 - Z_1)(Z_1 - Z_3)}{(Z_4 - Z_1)} \right|$ (b) $\left| \frac{(Z_2 - Z_1)(Z_3 - Z_1)}{(Z_4 - Z_1)} \right|$
 (c) $\left| \frac{(Z_2 - Z_1)(Z_3 - Z_1)}{(Z_4 - Z_1)^2} \right|$ (d) $\left| \frac{(Z_2 + Z_1)(Z_3 + Z_1)}{(Z_4 + Z_1)} \right|$

9. The value of $(Z_4 - Z_1)^2(1 + \cos \theta) \sec \theta$ is :

- (a) $(Z_2 - Z_1)(Z_3 - Z_1)$ (b) $\frac{(Z_2 - Z_1)(Z_3 - Z_1)}{Z_4 - Z_1}$
 (c) $\frac{(Z_2 - Z_1)(Z_3 - Z_1)}{(Z_4 - Z_1)^2}$ (d) $(Z_2 - Z_1)(Z_3 - Z_1)^2$

Answers

1.	(a)	2.	(b)	3.	(b)	4.	(d)	5.	(a)	6.	(d)	7.	(d)	8.	(c)	9.	(a)		
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Exercise-4 : Matching Type Problems

1. In a $\triangle ABC$, the side lengths BC , CA and AB are consecutive positive integers in increasing order.

Column-I		Column-II	
(A)	If z_1, z_2 and z_3 be the affixes of vertices A, B and C respectively in argand plane, such that $\left \arg \left(\frac{z_1 - z_3}{z_2 - z_3} \right) \right = \left 2 \arg \left(\frac{z_3 - z_1}{z_2 - z_1} \right) \right $, then biggest side of the triangle is	(P)	2
(B)	Let $\vec{a}, \vec{b}$ and $\vec{c}$ be the position vectors of vertices A, B and C respectively. If $(\vec{c} - \vec{a}) \cdot (\vec{b} - \vec{c}) = 0$ then the value of $ \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} $ equals to	(Q)	3
(C)	Let the equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ represent the lines AB and AC respectively and $\frac{a_1b_2 - a_2b_1}{a_1a_2 + b_1b_2} = \frac{4}{3}$ then the value of $s - c$ (where s is the semiperimeter) $a = BC, b = CA, c = AB$	(R)	4
(D)	If the altitudes of $\triangle ABC$ are in harmonic progression then the side length ' b ' can be	(S)	6
		(T)	12

2. Let $ABCDEF$ is a regular hexagon $A(z_1), B(z_2), C(z_3), D(z_4), E(z_5), F(z_6)$ in argand plane where A, B, C, D, E and F are taken in anticlockwise manner. If $z_1 = -2, z_3 = 1 - \sqrt{3}i$.

Column-I		Column-II	
(A)	If $z_2 = a + ib$, then $2a^2 + b^2$ is equal to	(P)	8
(B)	The square of the inradius of hexagon is	(Q)	7
(C)	The area of region formed by point $P(z)$ lying inside the incircle of hexagon and satisfying $\frac{\pi}{3} \leq \arg(z) \leq \frac{5\pi}{6}$ is $\frac{m}{n}\pi$, where m, n are relatively prime natural numbers, then $m + n$ is equal to	(R)	5
(D)	The value of $z_4^2 - z_1^2 - z_2^2 - z_3^2 - z_5^2 - z_6^2$ is equal to	(S)	3
		(T)	2

3.

Column-I		Column-II	
(A)	Let ω be a non real cube root of unity then the number of distinct elements in the set $\{(1 + \omega + \omega^2 + \dots + \omega^n)^m; n, m \in N\}$ is :	(P)	3
(B)	Let ω and ω^2 be non real cube root of unity. The least possible degree of a polynomial with real co-efficients having roots $2\omega, (2 + 3\omega), (2 + 3\omega)^2, (2 - \omega - \omega^2)$ is	(Q)	4
(C)	Let $\alpha = 6 + 4i$ and $\beta = 2 + 4i$ are two complex numbers on Argand plane. A complex number z satisfying $\arg\left(\frac{z - \alpha}{z - \beta}\right) = \frac{\pi}{6}$ moves on a major segment of a circle whose radius is	(R)	5
(D)	Let z_1, z_2, z_3 are complex numbers denoting the vertices of an equilateral triangle ABC having circumradius equals to unity. If P denotes any arbitrary point on its circumcircle then the value of $\frac{1}{2}((PA)^2 + (PB)^2 + (PC)^2)$ equals to	(S)	7

Answers

- $A \rightarrow S; B \rightarrow T; C \rightarrow S; D \rightarrow Q, R, S, T$
- $A \rightarrow R; B \rightarrow S; C \rightarrow Q; D \rightarrow P$
- $A \rightarrow S; B \rightarrow R; C \rightarrow Q; D \rightarrow P$

Exercise-5 : Subjective Type Problems

- Let complex number 'z' satisfy the inequality $2 \leq |z| \leq 4$. A point P is selected in this region at random. The probability that argument of P lies in the interval $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is $\frac{1}{K}$, then $K =$
- Let z be a complex number satisfying $|z-3| \leq |z-1|$, $|z-3| \leq |z-5|$, $|z-i| \leq |z+i|$ and $|z-i| \leq |z-5i|$. Then the area of region in which z lies is A square units, where $A =$
- Complex number z_1 and z_2 satisfy $z + \bar{z} = 2|z-1|$ and $\arg(z_1 - z_2) = \frac{\pi}{4}$. Then the value of $\operatorname{Im}(z_1 + z_2)$ is :
- If $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 36$, then $|z_1 + z_2 + z_3|$ is equal to :
- If $|z_1|$ and $|z_2|$ are the distances of points on the curve $5z\bar{z} - 2(z^2 - \bar{z}^2) - 9 = 0$ which are at maximum and minimum distance from the origin, then the value of $|z_1| + |z_2|$ is equal to :
- Let $\frac{1}{a_1 + \omega} + \frac{1}{a_2 + \omega} + \frac{1}{a_3 + \omega} + \dots + \frac{1}{a_n + \omega} = i$ where $a_1, a_2, a_3, \dots, a_n \in \mathbb{R}$ and ω is imaginary cube root of unity, then evaluate $\sum_{r=1}^n \frac{2a_r - 1}{a_r^2 - a_r + 1}$.
- If $|z_1| = 2$, $|z_2| = 3$, $|z_3| = 4$ and $|2z_1 + 3z_2 + 4z_3| = 9$, then value of $|8z_2z_3 + 27z_3z_1 + 64z_1z_2|^{1/3}$ is :
- The sum of maximum and minimum modulus of a complex number z satisfying $|z - 25i| \leq 15$, $i = \sqrt{-1}$ is S , then $\frac{S}{10}$ is :

Answers

1.	4	2.	6	3.	2	4.	6	5.	4	6.	0	7.	6
8.	5												

□□□

Exercise-1 : Single Choice Problems

- Let $A = BB^T + CC^T$, where $B = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$, $C = \begin{bmatrix} \sin \theta \\ -\cos \theta \end{bmatrix}$; $\theta \in R$. Then A is :
 (a) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
- Let $A = \begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$. The only correct statement about the matrix A is :
 (a) A is a zero matrix (b) $A^2 = I$, where I is a unit matrix
 (c) A^{-1} does not exist (d) $A = (-1)I$, where I is a unit matrix
- Let $A = [a_{ij}]_{3 \times 3}$ be such that $a_{ij} = \begin{cases} 3; & \text{when } i = j \\ 0; & i \neq j \end{cases}$, then $\left\{ \frac{\det(\text{adj}(\text{adj } A))}{5} \right\}$ equals :
 (where $\{ \cdot \}$ denotes fractional part function)
 (a) $\frac{2}{5}$ (b) $\frac{1}{5}$ (c) $\frac{2}{3}$ (d) $\frac{1}{3}$
- If $A^{-1} = \begin{bmatrix} \sin^2 \alpha & 0 & 0 \\ 0 & \sin^2 \beta & 0 \\ 0 & 0 & \sin^2 \gamma \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} \cos^2 \alpha & 0 & 0 \\ 0 & \cos^2 \beta & 0 \\ 0 & 0 & \cos^2 \gamma \end{bmatrix}$ where α, β, γ are any real numbers and $C = (A^{-5} + B^{-5}) + 5A^{-1}B^{-1}(A^{-3} + B^{-3}) + 10A^{-2}B^{-2}(A^{-1} + B^{-1})$ then find $|C|$.
 (a) 0 (b) 1 (c) 2 (d) 3
- If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$; then $A^{-1} =$
 (a) A (b) A^2 (c) A^3 (d) A^4
- Let $M = [a_{ij}]_{3 \times 3}$ where $a_{ij} \in \{-1, 1\}$. Find the maximum possible value of $\det(M)$.
 (a) 3 (b) 4 (c) 5 (d) 6

7. Let matrix $A = \begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$; if $xyz = 2\lambda$ and $8x + 4y + 3z = \lambda + 28$, then $(\text{adj } A)A$ equals :

(a) $\begin{bmatrix} \lambda+1 & 0 & 0 \\ 0 & \lambda+1 & 0 \\ 0 & 0 & \lambda+1 \end{bmatrix}$

(b) $\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$

(c) $\begin{bmatrix} \lambda^2 & 0 & 0 \\ 0 & \lambda^2 & 0 \\ 0 & 0 & \lambda^2 \end{bmatrix}$

(d) $\begin{bmatrix} \lambda+2 & 0 & 0 \\ 0 & \lambda+2 & 0 \\ 0 & 0 & \lambda+2 \end{bmatrix}$

8. If the trace of matrix $A = \begin{pmatrix} x-2 & e^x & -\sin x \\ \cos x^2 & x^2-x+3 & \ln|x| \\ 0 & \tan^{-1}x & x-7 \end{pmatrix}$ is zero, then x is equal to :

(a) -2 or 3

(b) -3 or -2

(c) -3 or 2

(d) 2 or 3

9. If $A = [a_{ij}]_{2 \times 2}$ where $a_{ij} = \begin{cases} i+j, & i \neq j \\ i^2-2j, & i = j \end{cases}$ then A^{-1} is equal to :

(a) $\frac{1}{9} \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$

(b) $\frac{1}{9} \begin{bmatrix} 0 & -3 \\ 3 & -1 \end{bmatrix}$

(c) $\frac{1}{9} \begin{bmatrix} 0 & -3 \\ -3 & -1 \end{bmatrix}$

(d) $\frac{1}{3} \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$

10. If $\begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta \\ -\tan \theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then :

(a) $a = b = 1$

(b) $a = \cos 2\theta, b = \sin 2\theta$

(c) $a = \sin 2\theta, b = \cos 2\theta$

(d) $a = 1, b = \sin 2\theta$

11. A square matrix P satisfies $P^2 = I - P$, where I is identity matrix. If $P^n = 5I - 8P$, then n is :

(a) 4

(b) 5

(c) 6

(d) 7

12. Let matrix $A = \begin{bmatrix} x & y & -z \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ where $x, y, z \in N$. If $\det. (\text{adj.} (\text{adj.} A)) = 2^8 \cdot 3^4$ then the number

of such matrices A is :

[Note : $\text{adj.} A$ denotes adjoint of square matrix A .]

(a) 220

(b) 45

(c) 55

(d) 110

13. If A is a 2×2 non singular matrix, then $\text{adj} (\text{adj } A)$ is equal to :

(a) A^2

(b) A

(c) A^{-1}

(d) $(A^{-1})^2$

14. $A = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$ and $MA = A^{2m}$, $m \in N$, $a, b \in R$, for some matrix M , then which one of the following is correct :

(a) $M = \begin{bmatrix} a^{2m} & b^{2m} \\ b^{2m} & -a^{2m} \end{bmatrix}$

(b) $M = (a^2 + b^2)^m \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(c) $M = (a^m + b^m) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(d) $M = (a^2 + b^2)^{m-1} \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$

- 15.** Let A be a square matrix satisfying $A^2 + 5A + 5I = 0$. The inverse of $A + 2I$ is equal to :
 (a) $A - 2I$ (b) $A + 3I$ (c) $A - 3I$ (d) non-existent
- 16.** Let $A = \begin{bmatrix} 3 & -5 \\ 7 & -12 \end{bmatrix}$ and $B = \begin{bmatrix} 12 & -5 \\ 7 & -3 \end{bmatrix}$ be two given matrices, then $(AB)^{-1}$ is :
 (a) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- 17.** If matrix $A = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}$ then the value of $|\text{adj. } A|$ equals to :
 (a) 2 (b) 3 (c) 4 (d) 6
- 18.** If for the matrix $A = \begin{bmatrix} \cos \theta & 2 \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, $A^{-1} = A^T$ then number of possible value(s) of θ in $[0, 2\pi]$ is:
 (a) 2 (b) 3 (c) 1 (d) 4
- 19.** Let M be a column vector (not null vector) and $A = \frac{MM^T}{M^T M}$ the matrix A is :
 (where M^T is transpose matrix of M)
 (a) idempotent (b) nilpotent (c) involutory (d) none of these
- 20.** If $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $P = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, $Q = P^T A P$, find $PQ^{2014}P^T$:
 (a) $\begin{pmatrix} 1 & 2^{2014} \\ 0 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 4028 \\ 0 & 1 \end{pmatrix}$
 (c) $(P^T)^{2013} A^{2014} P^{2013}$ (d) $P^T A^{2014} P$
- 21.** If M be a square matrix of order 3 such that $|M| = 2$, then $\left| \text{adj} \left(\frac{M}{2} \right) \right|$ equals to :
 (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{8}$ (d) $\frac{1}{16}$
- 22.** If A is matrix of order 3 such that $|A| = 5$ and $B = \text{adj } A$, then the value of $||A^{-1}|(AB)^T|$ is equal to
 (where $|A|$ denotes determinant of matrix A , A^T denotes transpose of matrix A , A^{-1} denotes inverse of matrix A , $\text{adj } A$ denotes adjoint of matrix A)
 (a) 5 (b) 1 (c) 25 (d) $\frac{1}{25}$

Answers

[illegible]

Exercise-2 : One or More than One Answer is/are Correct

- If A and B are two orthogonal matrices of order n and $\det(A) + \det(B) = 0$, then which of the following must be correct ?
 - $\det(A + B) = \det(A) + \det(B)$
 - $\det(A + B) = 0$
 - A and B both are singular matrices
 - $A + B = 0$
- Let M be a 3×3 matrix satisfying $M^3 = 0$. Then which of the following statement(s) are true:
 - $\left| \frac{1}{2}M^2 + M + I \right| \neq 0$
 - $\left| \frac{1}{2}M^2 - M + I \right| = 0$
 - $\left| \frac{1}{2}M^2 + M + I \right| = 0$
 - $\left| \frac{1}{2}M^2 - M + I \right| \neq 0$
- Let $A_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then :
 - $A_{\alpha+\beta} = A_\alpha A_\beta$
 - $A_\alpha^{-1} = A_{-\alpha}$
 - $A_\alpha^{-1} = -A_\alpha$
 - $A_\alpha^2 = -I$
- $A^3 - 2A^2 - A + 2I = 0$ if $A =$
 - I
 - $2I$
 - $\begin{bmatrix} 2 & -1 & 2 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$
 - $\begin{bmatrix} 2 & 1 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$
- Let A be a 3×3 symmetric invertible matrix with real positive elements. Then the number of zero elements in A^{-1} are less than or equal to :
 - 0
 - 1
 - 2
 - 3

Answers

1.	(a, b)	2.	(a, d)	3.	(a, b)	4.	(a, b, c, d)	5.	(d)
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Exercise-3 : Matching Type Problems

1. Consider a square matrix A of order 2 which has its elements as 0, 1, 2 and 4. Let N denotes the number of such matrices.

	Column-I		Column-II
(A)	Possible non-negative value of $\det(A)$ is	(P)	2
(B)	Sum of values of determinants corresponding to N matrices is	(Q)	4
(C)	If absolute value of $(\det(A))$ is least, then possible value of $ \text{adj}(\text{adj}(\text{adj } A)) $	(R)	-2
(D)	If $\det(A)$ is least, then possible value of $\det(4A^{-1})$ is	(S)	0
		(T)	8

2.

	Column-I		Column-II
(A)	If A is an idempotent matrix and I is an identify matrix of the same order, then the value of n , such that $(A + I)^n = I + 127A$ is	(P)	9
(B)	If $(I - A)^{-1} = I + A + A^2 + \dots + A^7$, then $A^n = O$ where n is	(Q)	10
(C)	If A is matrix such that $a_{ij} = (i + j)(i - j)$, then A is singular if order of matrix is	(R)	7
(D)	If a non-singular matrix A is symmetric, such that A^{-1} is also symmetric, then order of A can be	(S)	8

3.

	Column-I		Column-II
(A)	Number of ordered pairs (x, y) of real numbers satisfying $\sin x + \cos y = 0$, $\sin^2 x + \cos^2 y = \frac{1}{2}$, $0 < x < \pi$ and $0 < y < \pi$, is equal to	(P)	0
(B)	Given $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three vectors such that $\vec{b}$ and $\vec{c}$ are unit like vectors and $ \vec{a} = 4$. If $\vec{a} + \lambda \vec{c} = 2\vec{b}$ then the sum of all possible values of λ is equal to	(Q)	2

(C)	If $P = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$, $10Q = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & t \\ 1 & -2 & 3 \end{bmatrix}$ and $Q = P^{-1}$, then the value of t is equal to	(R)	4
(D)	If $y = \tan u$ where $u = v - \frac{1}{v}$ and $v = \ln x$, then the value of $\frac{dy}{dx}$ at $x = e$ is equal to λ then $[\lambda]$ is equal to (where $[\cdot]$ denotes greatest integer function)	(S)	5

4.

Column-I		Column-II	
(A)	If P and Q are variable points on $C_1 : x^2 + y^2 = 4$ and $C_2 : x^2 + y^2 - 8x - 6y + 24 = 0$ respectively then the maximum value of PQ is equal to	(P)	1
(B)	Let P, Q, R be invertible matrices of second order such that $A = PQ^{-1}, B = QR^{-1}, C = RP^{-1}$, then the value of $\det. (ABC + BCA + CAB)$ is equal to	(Q)	2
(C)	The perpendicular distance of the point whose position vector is $(1, 3, 5)$ from the line $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ is equal to	(R)	9
(D)	Let $f(x)$ be a continuous function in $[-1, 1]$ such that $f(x) = \begin{cases} \frac{\ln(px^2 + qx + r)}{x^2} & ; -1 \leq x < 0 \\ 1 & ; x = 0 \\ \frac{\sin(e^{x^2} - 1)}{x^2} & ; 0 < x \leq 1 \end{cases}$ then the value of $(p + q + r)$, is equal to	(S)	8

5.

Column-I		Column-II	
(A)	$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} \right)$ has the value equal to	(P)	1

(B)	Let $A = [a_{ij}]$ be a 3×3 matrix where $a_{ij} = \begin{cases} 2 \cos t; & \text{if } i = j \\ 1; & \text{if } i - j = 1 \\ 0; & \text{otherwise} \end{cases}$ then maximum value of $\det(A)$ is	(Q)	2
(C)	Let $f(x) = x^3 + px^2 + qx + 6$; where $p, q \in R$ and $f'(x) < 0$ in largest possible interval $\left(-\frac{5}{3}, -1\right)$ then value of $q - p$ is	(R)	3
(D)	If $4^x - 2^{x+2} + 5 + b - 1 - 3 = \sin y$; $x, y, b \in R$ then the sum of the possible values of b is λ then $(\lambda + 1)$ equals	(S)	4

Answers

1. $A \rightarrow P, Q, T; B \rightarrow S; C \rightarrow P, R; D \rightarrow R$
2. $A \rightarrow R; B \rightarrow P, Q, S; C \rightarrow P, R; D \rightarrow P, Q, R, S$
3. $A \rightarrow Q; B \rightarrow R; C \rightarrow S; D \rightarrow P$
4. $A \rightarrow S; B \rightarrow R; C \rightarrow P; D \rightarrow Q$
5. $A \rightarrow Q; B \rightarrow S; C \rightarrow P; D \rightarrow R$

Exercise-4 : Subjective Type Problems

1. A and B are two square matrices. Such that $A^2B = BA$ and if $(AB)^{10} = A^k \cdot B^{10}$. Find the value of $k - 1020$.

2. Let A_n and B_n be square matrices of order 3, which are defined as :

$$A_n = [a_{ij}] \text{ and } B_n = [b_{ij}] \text{ where } a_{ij} = \frac{2i+j}{3^{2n}} \text{ and } b_{ij} = \frac{3i-j}{2^{2n}} \text{ for all } i \text{ and } j, 1 \leq i, j \leq 3.$$

If $l = \lim_{n \rightarrow \infty} \text{Tr.} (3A_1 + 3^2A_2 + 3^3A_3 + \dots + 3^nA_n)$ and

$$m = \lim_{n \rightarrow \infty} \text{Tr.} (2B_1 + 2^2B_2 + 2^3B_3 + \dots + 2^nB_n), \text{ then find the value of } \frac{(l+m)}{3}$$

[Note : Tr. (P) denotes the trace of matrix P]

3. Let A be a 2×3 matrix whereas B be a 3×2 matrix. If $\det. (AB) = 4$, then the value of $\det. (BA)$, is :

4. Find the maximum value of the determinant of an arbitrary 3×3 matrix A , each of whose entries $a_{ij} \in \{-1, 1\}$.

5. The set of natural numbers is divided into array of rows and columns in the form of matrices as

$$A_1 = [1], A_2 = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}, A_3 = \begin{bmatrix} 6 & 7 & 8 \\ 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix} \text{ and so on. Let the trace of } A_{10} \text{ be } \lambda. \text{ Find unit digit of } \lambda?$$

Answers

1.	3	2.	7	3.	0	4.	4	5.	5				
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Exercise-1 : Single Choice Problems

- The number of 3-digit numbers containing the digit 7 exactly once :
 (a) 225 (b) 220 (c) 200 (d) 180
- Let $A = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8\}$, $B = \{y_1, y_2, y_3, y_4\}$. The total number of function $f : A \rightarrow B$ that are onto and there are exactly three elements x in A such that $f(x) = y_1$ is :
 (a) 11088 (b) 10920 (c) 13608 (d) None of these
- The number of arrangements of the word "IDIOTS" such that vowels are at the places which form three consecutive terms of an A.P. is :
 (a) 36 (b) 72 (c) 24 (d) 108
- Consider all the 5 digit numbers where each of the digits is chosen from the set $\{1, 2, 3, 4\}$. Then the number of numbers, which contain all the four digits is :
 (a) 240 (b) 244 (c) 586 (d) 781
- How many ways are there to arrange the letters of the word "GARDEN" with the vowels in alphabetical order ?
 (a) 120 (b) 480 (c) 360 (d) 240
- If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$ then the equation having α/β and β/α as its roots is :
 (a) $3x^2 - 19x + 3 = 0$ (b) $3x^2 + 19x - 3 = 0$
 (c) $3x^2 - 19x - 3 = 0$ (d) $x^2 - 5x + 3 = 0$
- A student is to answer 10 out of 13 questions in an examination such that he must choose at least 4 from the first five questions. The number of choices available to him is :
 (a) 140 (b) 196 (c) 280 (d) 346
- Let set $A = \{1, 2, 3, \dots, 22\}$. Set B is a subset of A and B has exactly 11 elements, find the sum of elements of all possible subsets B .
 (a) $252 {}^{21}C_{11}$ (b) $230 {}^{21}C_{10}$
 (c) $253 {}^{21}C_9$ (d) $253 {}^{21}C_{10}$

9. The value of $\left[\frac{2009! + 2006!}{2008! + 2007!} \right] =$
 ([.] denotes greatest integer function.)
 (a) 2009 (b) 2008 (c) 2007 (d) 1
10. If $p_1, p_2, p_3, \dots, p_{m+1}$ are distinct prime numbers. Then the number of factors of $p_1^n p_2 p_3 \dots p_{m+1}$ is :
 (a) $m(n+1)$ (b) $(n+1)2^m$ (c) $n \cdot 2^m$ (d) 2^{nm}
11. A basket ball team consists of 12 pairs of twin brothers. On the first day of training, all 24 players stand in a circle in such a way that all pairs of twin brothers are neighbours. Number of ways this can be done is :
 (a) $(12)! 2^{11}$ (b) $(11)! 2^{12}$ (c) $(12)! 2^{12}$ (d) $(11)! 2^{11}$
12. Let 'm' denotes the number of four digit numbers such that the left most digit is odd, the second digit is even and all four digits are different and 'n' denotes the number of four digit numbers such that left most digit is even, second digit is odd and all four digits are different. If $m = nk$, then k equals :
 (a) $\frac{4}{5}$ (b) $\frac{3}{4}$ (c) $\frac{5}{4}$ (d) $\frac{4}{3}$
13. The number of three digit numbers of the form xyz such that $x < y$ and $z \leq y$ is :
 (a) 156 (b) 204 (c) 240 (d) 276
14. A and B are two sets and their intersection has 3 elements. If A has 1920 more subsets than B has, then the number of elements of A union B is :
 (a) 12 (b) 14 (c) 15 (d) 16
15. All possible 120 permutations of WDSMC are arranged in dictionary order, as if each were an ordinary five-letter word. The last letter of the 86th word in the list, is :
 (a) W (b) D (c) M (d) C
16. The number of permutation of all the letters AAAABBBBC in which the A's appear together in a block of 4 letters or the B's appear together in a block of 3 letters is :
 (a) 44 (b) 50 (c) 60 (d) 89
17. Number of zero's at the ends of $\prod_{n=5}^{30} (n)^{n+1}$ is :
 (a) 111 (b) 147 (c) 137 (d) None of these
18. The number of positive integral pairs (x, y) satisfying the equation $x^2 - y^2 = 3370$ is :
 (a) 0 (b) 1 (c) 2 (d) 4
19. The number of ways of selecting 'n' things out of '3n' things of which 'n' are of one kind and alike and 'n' are of second kind and alike and the rest unlike is :
 (a) $n 2^{n-1}$ (b) $(n-1) 2^{n-1}$ (c) $(n+1) 2^{n-1}$ (d) $(n+2) 2^{n-1}$

20. If x, y, z are three natural numbers in A.P. such that $x + y + z = 30$, then the possible number of ordered triplet (x, y, z) is :
 (a) 18 (b) 19 (c) 20 (d) 21
21. A dice is rolled 4 times, the numbers appearing are listed. The number of different throws, such that the largest number appearing in the list is not 4, is :
 (a) 175 (b) 625 (c) 1040 (d) 1121
22. Let m denotes the number of ways in which 5 boys and 5 girls can be arranged in a line alternately and n denotes the number of ways in which 5 boys and 5 girls can be arranged in a circle so that no two boys are together. if $m = kn$ then the value of k is :
 (a) 2 (b) 5 (c) 6 (d) 10
23. Number of ways in which 4 students can sit in 7 chair in a row, if there is no empty chair between any two students is :
 (a) 24 (b) 28 (c) 72 (d) 96
24. Number of zero's at the ends of $\prod_{n=5}^{30} (n)^{n+1}$ is :
 (a) 111 (b) 147 (c) 137 (d) None
25. The number of words of four letters consisting of equal number of vowels and consonants (of english language) with repetition permitted is :
 (a) 51030 (b) 50030 (c) 63050 (d) 66150
26. Ten different letters of an alphabet are given. Words with five letters are formed with these given letters. Then the number of words which have atleast one letter repeated is :
 (a) 30240 (b) 69760 (c) 69780 (d) 99784
27. Number of four digit numbers in which at least one digit occurs more than once, is:
 (a) 4464 (b) 4644 (c) 4446 (d) 6444
28. In a game of minesweeper, a number on a square denotes the number of mines that share at least one vertex with that square. A square with a number may not have a mine, and the blank squares are undetermined. In how many ways can the mines be placed in the given configuration on blank squares:

	2		1		2

- (a) 120 (b) 105 (c) 95 (d) 100
29. Let the product of all the divisors of 1440 be P . If P is divisible by 24^x , then the maximum value of x is :
 (a) 28 (b) 30 (c) 32 (d) 36

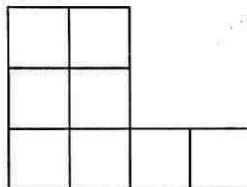
30. Let N be the number of 4-digit numbers which contain not more than 2 different digits. The sum of the digits of N is :
 (a) 18 (b) 19 (c) 20 (d) 21
31. The number of different permutations of all the letters of the word PERMUTATION such that any two consecutive letters in the arrangement are neither both vowels nor both identical is :
 (a) $63 \times \underline{6} \times \underline{5}$ (b) $8 \times \underline{6} \times \underline{5}$ (c) $57 \times \underline{5} \times \underline{5}$ (d) $7 \times \underline{7} \times \underline{5}$
32. A batsman can score 0, 1, 2, 3, 4 or 6 runs from a ball. The number of different sequences in which he can score exactly 30 runs in an over of six balls :
 (a) 4 (b) 72 (c) 56 (d) 71
33. A batsman can score 0, 2, 3, or 4 runs for each ball he receives. If N is the number of ways of scoring a total of 20 runs in one over of six balls, then N is divisible by:
 (a) 5 (b) 7 (c) 14 (d) 16
34. The number of non-negative integral solutions of the equation $x + y + z = 5$ is :
 (a) 20 (b) 19 (c) 21 (d) 25
35. The number of solutions of the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 101$, where x_i 's are odd natural numbers is :
 (a) $^{105}C_4$ (b) $^{52}C_5$ (c) $^{52}C_4$ (d) $^{50}C_4$
36. An ordinary dice is rolled 4 times, numbers appearing on them are listed. The number of different throws, such that the largest number appearing on them is NOT 4, is :
 (a) 175 (b) 625 (c) 1121 (d) 1040
37. Number of four letter words can be formed using the letters of word VIBRANT if letter V is must included, are :
 (a) 840 (b) 480 (c) 120 (d) 240
38. The number of rectangles that can be obtained by joining four of the twelve vertices of a 12-sided regular polygon is :
 (a) 66 (b) 30 (c) 24 (d) 15
39. Number of five digit integers, with sum of the digits equal to 43 are :
 (a) 5 (b) 10 (c) 15 (d) 35

Answers

1. (a)	2. (d)	3. (d)	4. (a)	5. (c)	6. (a)	7. (b)	8. (d)	9. (b)	10. (b)
11. (b)	12. (c)	13. (d)	14. (c)	15. (b)	16. (a)	17. (c)	18. (a)	19. (d)	20. (b)
21. (d)	22. (d)	23. (d)	24. (c)	25. (d)	26. (b)	27. (a)	28. (c)	29. (b)	30. (a)
31. (c)	32. (d)	33. (d)	34. (c)	35. (c)	36. (c)	37. (b)	38. (d)	39. (c)	

Exercise-2 : One or More than One Answer is/are Correct

- The number of 5 letter words formed with the letters of the word CALCULUS is divisible by :
 (a) 2 (b) 3 (c) 5 (d) 7
- The coefficient of x^{50} in the expansion of $\sum_{k=0}^{100} {}^{100}C_k (x-2)^{100-k} 3^k$ is also equal to :
 (a) Number of ways in which 50 identical books can be distributed in 100 students, if each student can get atmost one book.
 (b) Number of ways in which 100 different white balls and 50 identical red balls can be arranged in a circle, if no two red balls are together.
 (c) Number of dissimilar terms in $(x_1 + x_2 + x_3 + \dots + x_{50})^{51}$.
 (d) $\frac{2 \cdot 6 \cdot 10 \cdot 14 \dots 198}{50!}$
- Number of ways in which the letters of the word "NATION" can be filled in the given figure such that no row remains empty and each box contains not more than one letter, are :



- (a) 11 | 6 (b) 12 | 6 (c) 13 | 6 (d) 14 | 6
- Let a, b, c, d be non zero distinct digits. The number of 4 digit numbers $abcd$ such that $ab + cd$ is even is divisible by :
 (a) 3 (b) 4 (c) 7 (d) 11

Answers

1.	(a, b, c)	2.	(a, d)	3.	(c)	4.	(a, b, d)				
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**Exercise-4 : Matching Type Problems**

1. All letters of the word BREAKAGE are to be jumbled. The number of ways of arranging them so that :

Column-I		Column-II	
(A)	The two A's are not together	(P)	720
(B)	The two E's are together but not two A's	(Q)	1800
(C)	Neither two A's nor two E's are together	(R)	5760
(D)	No two vowels are together	(S)	6000
		(T)	7560

2. Consider the letters of the word MATHEMATICS. Set of repeating letters = { M, A, T }, set of non repeating letters = { H, E, I, C, S }:

Column-I		Column-II	
(A)	The number of words taking all letters of the given word such that atleast one repeating letter is at odd position is	(P)	$28 \cdot (7!)$
(B)	The number of words formed taking all letters of the given word in which no two vowels are together is	(Q)	$\frac{(11)!}{(2!)^3}$
(C)	The number of words formed taking all letters of the given word such that in each word both M's are together and both T's are together but both A's are not together is	(R)	$210(7!)$
(D)	The number of words formed taking all letters of the given word such that relative order of vowels and consonants does not change is	(S)	$840 (7!)$
		(T)	$\frac{4!7!}{(2!)^3}$

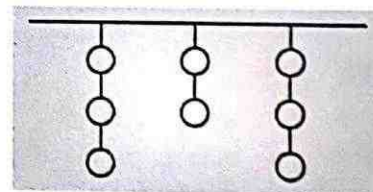
Answers

1. A $\rightarrow$ T; B $\rightarrow$ Q; C $\rightarrow$ R; D $\rightarrow$ P

2. A $\rightarrow$ Q; B $\rightarrow$ R; C $\rightarrow$ P; D $\rightarrow$ T


Exercise-5 : Subjective Type Problems

1. The number of ways in which eight digit number can be formed using the digits from 1 to 9 without repetition if first four places of the numbers are in increasing order and last four places are in decreasing order is N , then find the value of $\frac{N}{70}$.
2. Number of ways in which the letters of the word DECISIONS be arranged so that letter N be somewhere to the right of the letter "D" is $\frac{9}{\lambda}$. Find λ .
4. There are 10 stations enroute. A train has to be stopped at 3 of them. Let N be the ways in which the train can be stopped if atleast two of the stopping stations are consecutive. Find the value of $\sqrt{N}$.
5. There are 10 girls and 8 boys in a class room including Mr. Ravi, Ms. Rani and Ms. Radha. A list of speakers consisting of 8 girls and 6 boys has to be prepared. Mr. Ravi refuses to speak if Ms. Rani is a speaker. Ms. Rani refuses to speak if Ms. Radha is a speaker. The number of ways the list can be prepared is a 3 digit number $n_1 n_2 n_3$, then $|n_3 + n_2 - n_1| =$
6. Nine people sit around a round table. The number of ways of selecting four of them such that they are not from adjacent seats, is
7. Let the number of arrangements of all the digits of the numbers 12345 such that atleast 3 digits will not come in it's original position is N . Then the unit digit of N is
8. The number of triangles with each side having integral length and the longest side is of 11 units is equal to k^2 , then the value of ' k ' is equal to
9. 8 clay targets are arranged as shown. If N be the number of different ways they can be shot (one at a time) if no target can be shot until the target(s) below it have been shot. Find the ten's digit of N .



10. There are n persons sitting around a circular table. They start singing a 2 minute song in pairs such that no two persons sitting together will sing together. This process is continued for 28 minutes. Find n .
11. The number of ways to choose 7 distinct natural numbers from the first 100 natural numbers such that any two chosen numbers differ atleast by 7 can be expressed as nC_7 . Find the number of divisors of n .
12. Four couples (husband and wife) decide to form a committee of four members. The number of different committees that can be formed in which no couple finds a place is λ , then the sum of digits of λ is :

- 13.** The number of ways in which $2n$ objects of one type, $2n$ of another type and $2n$ of a third type can be divided between 2 persons so that each may have $3n$ objects is $\alpha n^2 + \beta n + \gamma$. Find the value of $(\alpha + \beta + \gamma)$.
- 14.** Let N be the number of integral solution of the equation $x + y + z + w = 15$ where $x \geq 0, y > 5, z \geq 2$ and $w \geq 1$. Find the unit digit of N .

Answers

1.	9	2.	8	3.	8	4.	8	5.	5	6.	9	7.	9
8.	6	9.	6	10.	7	11.	7	12.	7	13.	7	14.	4



Exercise-1 : Single Choice Problems

- Let $N = 2^{1224} - 1$, $\alpha = 2^{153} + 2^{77} + 1$ and $\beta = 2^{408} - 2^{204} + 1$. Then which of the following statement is correct ?
 (a) α divides N but β does not
 (b) β divides N but α does not
 (c) α and β both divide N
 (d) neither α nor β divides N
- If $(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then $a_r - {}^nC_1 \cdot a_{r-1} + {}^nC_2 a_{r-2} - {}^nC_3 a_{r-3} + \dots + (-1)^r {}^nC_r a_0$ is equal to : (r is not multiple of 3)
 (a) 0
 (b) nC_r
 (c) a_r
 (d) 1
- The coefficient of the middle term in the binomial expansion in powers of x of $(1 + \alpha x)^4$ and of $(1 - \alpha x)^6$ is the same if α equals :
 (a) $-\frac{5}{3}$
 (b) $\frac{3}{5}$
 (c) $-\frac{3}{10}$
 (d) $\frac{10}{3}$
- If $(1 + x)^{2010} = C_0 + C_1 x + C_2 x^2 + \dots + C_{2010} x^{2010}$ then the sum of series $C_2 + C_5 + C_8 + \dots + C_{2009}$ equals to :
 (a) $\frac{1}{2}(2^{2010} - 1)$
 (b) $\frac{1}{3}(2^{2010} - 1)$
 (c) $\frac{1}{2}(2^{2009} - 1)$
 (d) $\frac{1}{3}(2^{2009} - 1)$
- Let $\alpha_n = (2 + \sqrt{3})^n$. Find $\lim_{n \rightarrow \infty} (\alpha_n - [\alpha_n])$ ($[\cdot]$ denotes greatest integer function)
 (a) 1
 (b) $\frac{1}{2}$
 (c) $\frac{1}{3}$
 (d) $\frac{2}{3}$
- The number $N = {}^{20}C_7 - {}^{20}C_8 + {}^{20}C_9 - {}^{20}C_{10} + \dots - {}^{20}C_{20}$ is not divisible by :
 (a) 3
 (b) 7
 (c) 11
 (d) 19

7. The value of the expression $\log_2 \left(1 + \frac{1}{2} \sum_{k=1}^{11} {}^{12}C_k \right)$:
- (a) 11 (b) 12 (c) 13 (d) 14
8. The constant term in the expansion of $\left(x + \frac{1}{x^3} \right)^{12}$ is :
- (a) 26 (b) 169 (c) 260 (d) 220
9. If $\frac{3}{4!} + \frac{4}{5!} + \frac{5}{6!} + \dots + 50$ term $= \frac{1}{3!} - \frac{1}{(k+3)!}$, then sum of coefficients in the expansion $(1 + 2x_1 + 3x_2 + \dots + 100x_{100})^k$ is :
- (where $x_1, x_2, x_3, \dots, x_{100}$ are independent variables)
- (a) $(5050)^{49}$ (b) $(5050)^{51}$
 (c) $(5050)^{52}$ (d) $(5050)^{50}$
10. **Statement-1:** The remainder when $(128)^{(128)^{128}}$ is divided by 7 is 3.
because
Statement-2: $(128)^{128}$ when divided by 3 leaves the remainder 1.
- (a) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (b) Statement-1 is true, statement-2 is true and statement-2 is not the correct explanation for statement-1.
 (c) Statement-1 is true, statement-2 is false.
 (d) Statement-1 is false, statement-2 is true.
11. If $n > 3$, then $xyz {}^nC_0 - (x-1)(y-1)(z-1) {}^nC_1 + (x-2)(y-2)(z-2) {}^nC_2 - (x-3)(y-3)(z-3) {}^nC_3 + \dots + (-1)^n (x-n)(y-n)(z-n) {}^nC_n$ equals :
- (a) xyz (b) $x+y+z$
 (c) $xy+yz+zx$ (d) 0
12. If $\alpha_1, \alpha_2, \dots, \alpha_n$ are the $n; n^{\text{th}}$ roots of unity, $\alpha_r = e^{\frac{i2(r-1)\pi}{n}}$, $r = 1, 2, \dots, n$ then ${}^nC_1 \alpha_1 + {}^nC_2 \alpha_2 + \dots + {}^nC_n \alpha_n$ is equal to:
- (a) $\left(1 + \frac{\alpha_2}{\alpha_1} \right)^n - 1$ (b) $\frac{\alpha_1}{2} [(1 + \alpha_1)^n - 1]$ (c) $\frac{\alpha_1 + \alpha_{n-1} - 1}{2}$ (d) $(\alpha_1 + \alpha_{n-1})^n - 1$
13. The remainder when $2^{30} \cdot 3^{20}$ is divided by 7 is :
- (a) 1 (b) 2 (c) 4 (d) 6
14. ${}^{26}C_0 + {}^{26}C_1 + {}^{26}C_2 + \dots + {}^{26}C_{13}$ is equal to :
- (a) $2^{25} - \frac{1}{2} \cdot {}^{26}C_{13}$ (b) $2^{25} + \frac{1}{2} \cdot {}^{26}C_{13}$ (c) 2^{13} (d) $2^{26} + \frac{1}{2} \cdot {}^{26}C_{13}$

15. If a_r is the coefficient of x^r in the expansion of $(1+x+x^2)^n$ ($n \in N$). Then the value of $(a_1 + 4a_4 + 7a_7 + 10a_{10} + \dots)$ is equal to :
 (a) 3^{n-1} (b) 2^n (c) $\frac{1}{3} \cdot 2^n$ (d) $n \cdot 3^{n-1}$
16. Let $\binom{n}{k}$ represents the combination of ' n ' things taken ' k ' at a time, then the value of the sum $\binom{99}{97} + \binom{98}{96} + \binom{97}{95} + \dots + \binom{3}{1} + \binom{2}{0}$ equals:
 (a) $\binom{99}{97}$ (b) $\binom{100}{98}$ (c) $\binom{99}{98}$ (d) $\binom{100}{97}$
17. The last digit of $9! + 3^{9966}$ is :
 (a) 1 (b) 3 (c) 7 (d) 9
18. Let x be the 7th term from the beginning and y be the 7th term from the end in the expansion of $\left(3^{1/3} + \frac{1}{4^{1/3}}\right)^n$. If $y = 12x$ then the value of n is :
 (a) 9 (b) 8 (c) 10 (d) 11
19. The expression $({}^{10}C_0)^2 - ({}^{10}C_1)^2 + ({}^{10}C_2)^2 - \dots + ({}^{10}C_8)^2 - ({}^{10}C_9)^2 + ({}^{10}C_{10})^2$ equals :
 (a) $10!$ (b) $({}^{10}C_5)^2$ (c) $-{}^{10}C_5$ (d) ${}^{10}C_5$
20. The ratio of the co-efficients to x^{15} to the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^{15}$ is :
 (a) 1 : 4 (b) 1 : 32 (c) 7 : 64 (d) 7 : 16
21. In the expansion of $(1+x)^2(1+y)^3(1+z)^4(1+w)^5$, the sum of the coefficient of the terms of degree 12 is :
 (a) 61 (b) 71 (c) 81 (d) 91
22. If $\sum_{r=0}^n \left(\frac{r^3 + 2r^2 + 3r + 2}{(r+1)^2} \right) {}^nC_r = \frac{2^4 + 2^3 + 2^2 - 2}{3}$
 then the value of n is :
 (a) 2 (b) 2^2 (c) 2^3 (d) 2^4

Answers

1. (c)	2. (a)	3. (c)	4. (b)	5. (a)	6. (c)	7. (a)	8. (d)	9. (d)	10. (d)
11. (d)	12. (a)	13. (b)	14. (b)	15. (d)	16. (d)	17. (d)	18. (a)	19. (c)	20. (b)
21. (d)	22. (a)								

Exercise-2 : One or More than One Answer is/are Correct

- The number $N = {}^{20}C_7 - {}^{20}C_8 + {}^{20}C_9 - {}^{20}C_{10} + \dots - {}^{20}C_{20}$ is divisible by :
 (a) 3 (b) 4 (c) 7 (d) 19
- If $(1 + x + x^2 + x^3)^{100} = a_0 + a_1x + a_2x^2 + \dots + a_{300}x^{300}$ then which of the following statement(s) is/are correct ?
 (a) $a_1 = 100$
 (b) $a_0 + a_1 + a_2 + \dots + a_{300}$ is divisible by 1024
 (c) coefficients equidistant from beginning and end are equal
 (d) $a_0 + a_2 + a_4 + \dots + a_{300} = a_1 + a_3 + a_5 + \dots + a_{299}$
- $\sum_{r=0}^4 (-1)^r {}^{16}C_r$ is divisible by :
 (a) 5 (b) 7 (c) 11 (d) 13
- The expansion of $\left(\sqrt{x} + \frac{1}{2\sqrt[4]{x}}\right)^n$ is arranged in decreasing powers of x . If coefficient of first three terms form an A.P. then in expansion, the integral powers of x are :
 (a) 0 (b) 2 (c) 4 (d) 8
- Let $(1 + x^2)^2(1 + x)^n = \sum_{k=0}^{n+4} a_k x^k$. If a_1, a_2, a_3 are in A.P. then n is (given that ${}^nC_r = 0$, if $n < r$):
 (a) 6 (b) 4 (c) 3 (d) 2
- $\sum_{i=0}^n \sum_{j=0}^n \sum_{k=0}^n \binom{n}{i} \binom{n}{j} \binom{n}{k}, \binom{n}{r} = {}^nC_r$:
 (a) is less than 500 if $n = 3$ (b) is greater than 600 if $n = 3$
 (c) is less than 5000 if $n = 4$ (d) is greater than 4000 if $n = 4$
- If ${}^{100}C_6 + 4 \cdot {}^{100}C_7 + 6 \cdot {}^{100}C_8 + 4 \cdot {}^{100}C_9 + {}^{100}C_{10}$ has the value equal to xC_y ; then the possible value(s) of $x + y$ can be :
 (a) 112 (b) 114 (c) 196 (d) 198
- If the co-efficient of x^{2r} is greater than half of the co-efficient of x^{2r+1} in the expansion of $(1 + x)^{15}$; then the possible value of ' r ' equal to :
 (a) 5 (b) 6 (c) 7 (d) 8
- Let $f(x) = 1 + x^{111} + x^{222} + x^{333} + \dots + x^{999}$ then $f(x)$ is divisible by
 (a) $x + 1$ (b) x
 (c) $x - 1$ (d) $1 + x^{222} + x^{444} + x^{666} + x^{888}$

Answers

1.	(a, b, c, d)	2.	(a, b, c, d)	3.	(a, b, d)	4.	(a, c, d)	5.	(b, c, d)	6.	(c, d)
7.	(b, d)	8.	(a, b, c)	9.	(a, d)						